

# A \$12 million battery. A \$57 million project.

Hyde Park BESS (Project 21334): where the cost of a utility-owned, distribution-connected battery storage station actually sits, and how it compares to industry benchmarks

Cost Estimating Center of Excellence | Estimate basis: ESF Total Rev 3, 24 Sep 2026 | 5 Oct 2026

Internal. Contains vendor-confidential pricing.

## THE PROMISE

# By the end of this deck you can answer one question in one breath

“Why is a \$12 million battery a \$57 million project?” Because a vendor prices the battery. We build the station, the site, the feeder protection and the siting case, and we carry the risk and the cost of money.

## \$12.1M

### Vendor fixed-price contract

Nomad cover agreement, 10 Mar 2026.  
12 Voyager LFP units, PCS, BMS, fire protection, 10-yr warranty. \$605/kWh, \$1,211/kW.

## \$57.2M

### Project 21334 approval level

ESF Total Rev 3. Station 360 \$50.0M plus Station 496 protection \$7.3M. \$2,862/kWh, \$5,725/kW. Conceptual, -25%/+50%.

## 21%

### Battery share of the total

Add the D-Line circuit extension (24211, \$4.8M conceptual) and the share drops to 20%. Four dollars of every five are not the battery.

**Three sections:** where the money sits (thirteen layers, unit costs, benchmarks, six price curves); how the cost changed (2021 to today); can the number be defended (red team, stress tests, what must close).

**Against 35 public projects its own size:** the battery installed sits at the 57% percentile, the station complete at the 77% (68% among the 25 utility-owned), and the approval level above every member. The gap is the station, the interconnection and the loaders, not the battery.

# What a vendor quote contains, and what it leaves for the utility

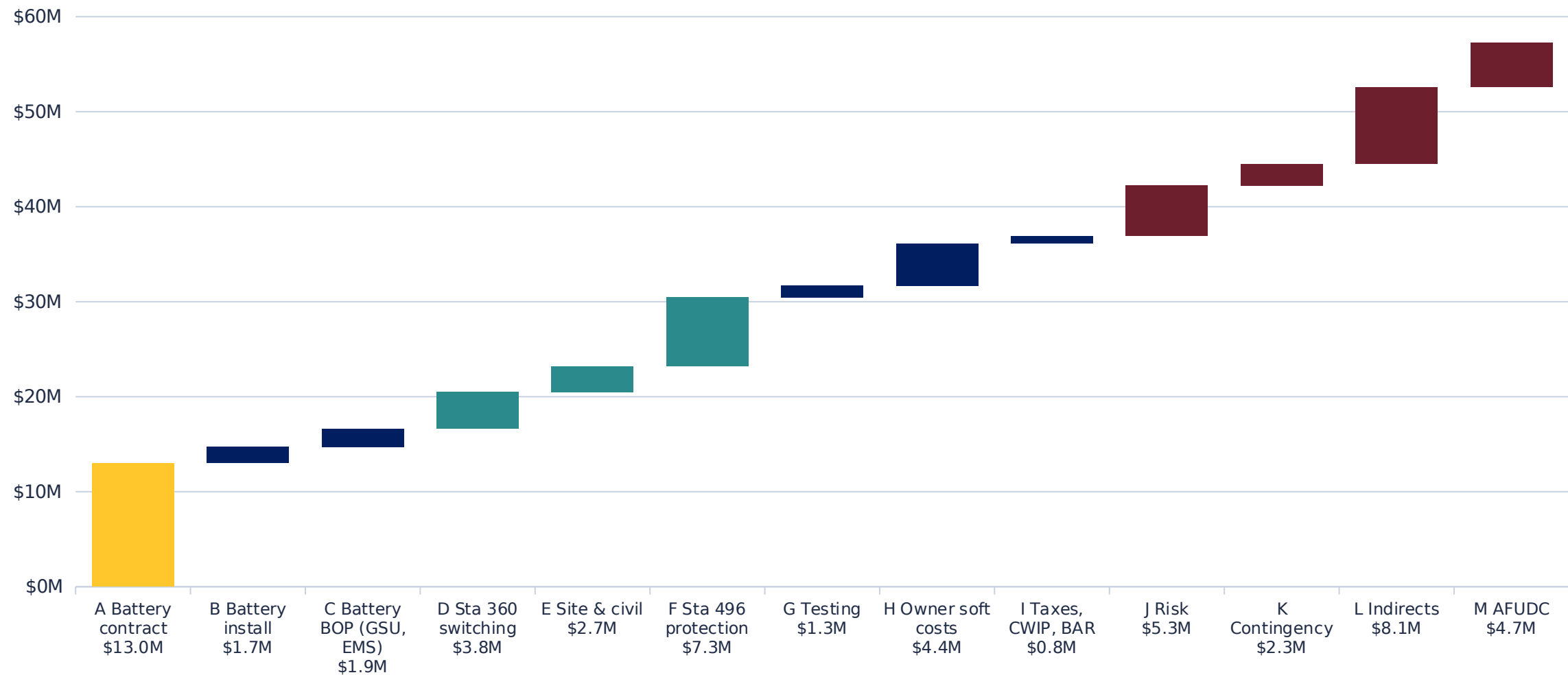
## Inside the \$12.1M Nomad price

- 12 containerized LFP units, 10 MW / 20 MWh, 12 MVA
- Inverters (PCS), BMS, HVAC, fire detection and suppression inside the container
- Spares \$125K, shipping \$90K, remote support \$210K
- 10-year warranty; years 3-7 extension \$798K
- Field assembly supervision, commissioning, training: \$246K
- Fixed until Final Acceptance; LDs capped at 10%

## Outside it: \$45.1M, 79% of the project

- **Station 360**, \$3.8M direct: four 15 kV breaker skids, GSUs, SSVT, metering, cable, duct bank, ground grid
- **Site**, \$2.7M direct: a 200 ft x 200 ft platform on a closed landfill, access road, retaining wall
- **Station 496**, \$7.3M: 20 relays, transfer trip, RTU work so four feeders can accept 2.5 MW each
- **Owner scope**, \$10.1M: install labor, GSUs, EMS, testing, engineering, Article 80 siting, outreach, PM, taxes
- **Station 360 risk, contingency, indirects, AFUDC**, \$20.3M: 35% of the total (Station 496 carries its own inside its \$7.3M)

# Thirteen layers between the battery contract and the approval level



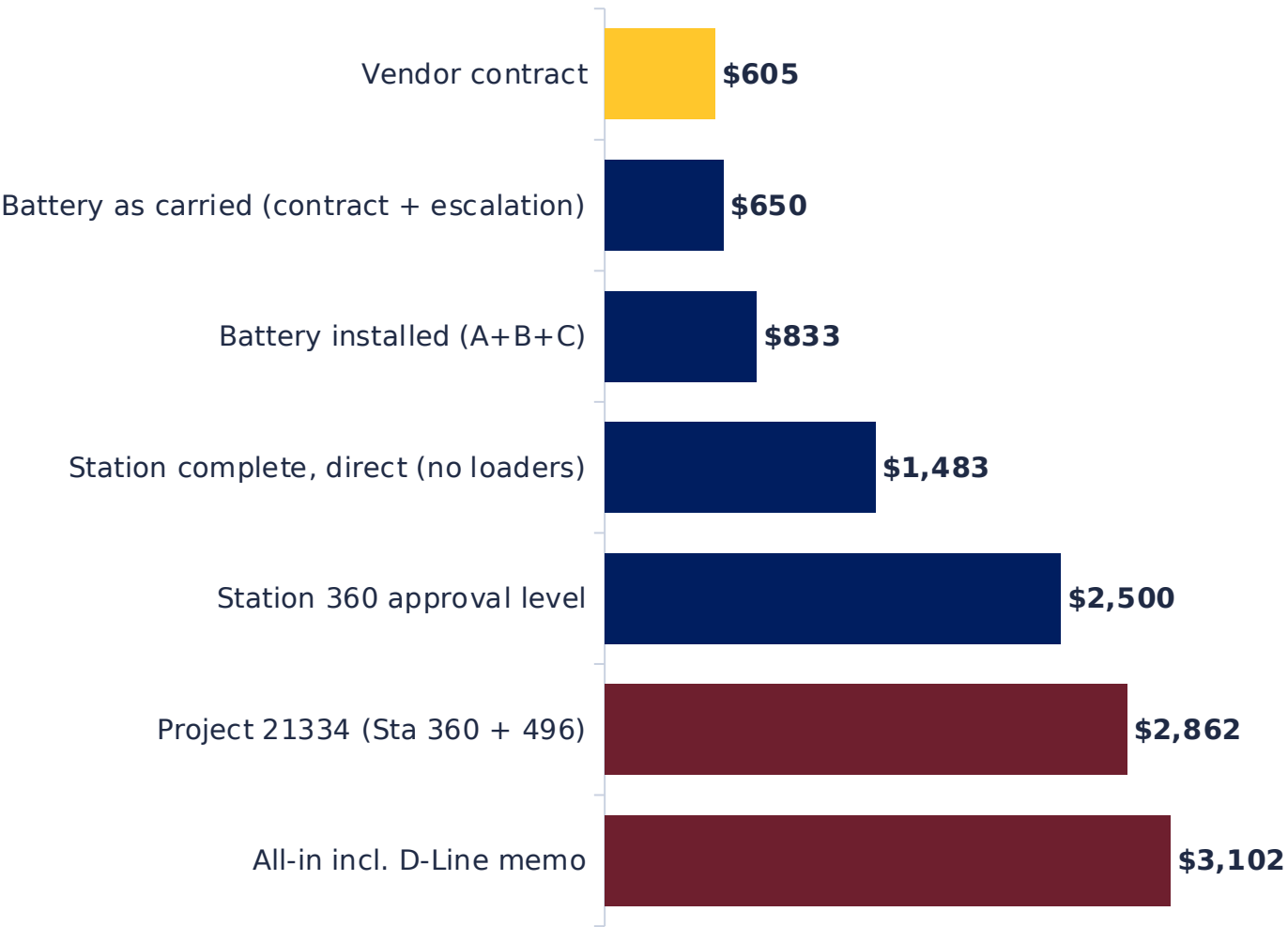
Gold = battery contract as carried in Rev 3 (\$12.1M contract plus \$1.1M escalation on the unpaid balance; see slide 23). Teal = station, site and feeder protection. Burgundy = risk, contingency, indirects, AFUDC. Total \$57.2M.  
EstimatingCoE | Eversource Energy | Stewards of Affordability

## Where the \$57.2M sits when risk, contingency, indirects and AFUDC are spread

| Layer                                                                     | Direct         | Loaded         | Share       |
|---------------------------------------------------------------------------|----------------|----------------|-------------|
| Battery supply contract (Nomad)                                           | \$13.0M        | \$21.9M        | 38%         |
| Battery installation & integration (owner scope)                          | \$1.7M         | \$2.9M         | 5%          |
| Battery balance-of-plant (GSUs, EMS/comms, aux power)                     | \$1.9M         | \$3.3M         | 6%          |
| Station 360 13.8 kV switching & interconnection facilities                | \$3.8M         | \$6.5M         | 11%         |
| Site development & civil (former landfill site)                           | \$2.7M         | \$4.5M         | 8%          |
| Station 496 feeder protection upgrades (interconnection)                  | \$7.3M         | \$7.3M         | 13%         |
| Testing & commissioning                                                   | \$1.3M         | \$2.1M         | 4%          |
| Owner soft costs (engineering, siting, environmental, outreach, PM, land) | \$4.4M         | \$7.4M         | 13%         |
| Taxes, CWIP, builders-risk insurance                                      | \$0.8M         | \$1.3M         | 2%          |
| <b>Total, project 21334 (directs incl. \$20.3M of loaders J-M)</b>        | <b>\$57.2M</b> | <b>\$57.2M</b> | <b>100%</b> |

Battery-related layers A+B+C loaded: \$28.1M (49%). Station, site and feeder interconnection D+E+F loaded: \$18.3M (32%). Owner soft costs, testing and taxes: \$10.9M.

# The same project is \$605/kWh or \$2,862/kWh depending on where you draw the line

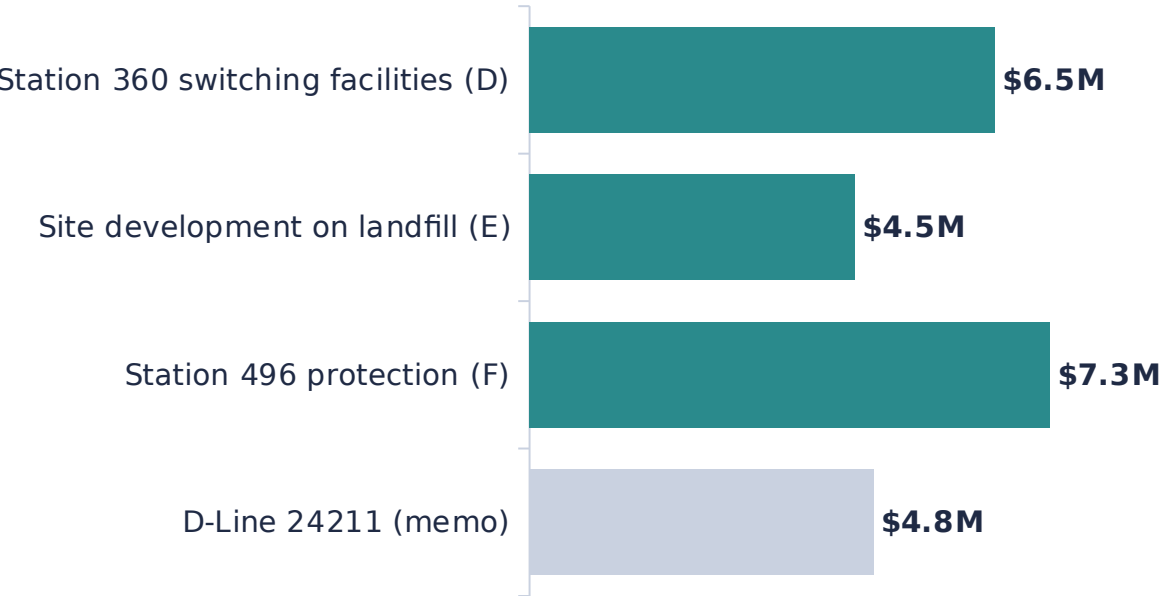


| Boundary          | \$/kW   | \$/kWh  | x    |
|-------------------|---------|---------|------|
| Vendor contract   | \$1,211 | \$605   | 1.0x |
| Battery installed | \$1,666 | \$833   | 1.4x |
| Sta 360 approval  | \$4,999 | \$2,500 | 4.1x |
| Project 21334     | \$5,725 | \$2,862 | 4.7x |

Incl. D-Line      \$6,204      \$3,102      5.1x

A benchmark is only meaningful against the row that matches its definition. Pack and DC-block prices belong beside the top bar; NREL and EIA installed costs beside 'battery installed' or 'station complete'; nothing published includes a utility's own indirects and AFUDC.

# Station and interconnection: \$18.3M loaded, none of it in a vendor quote

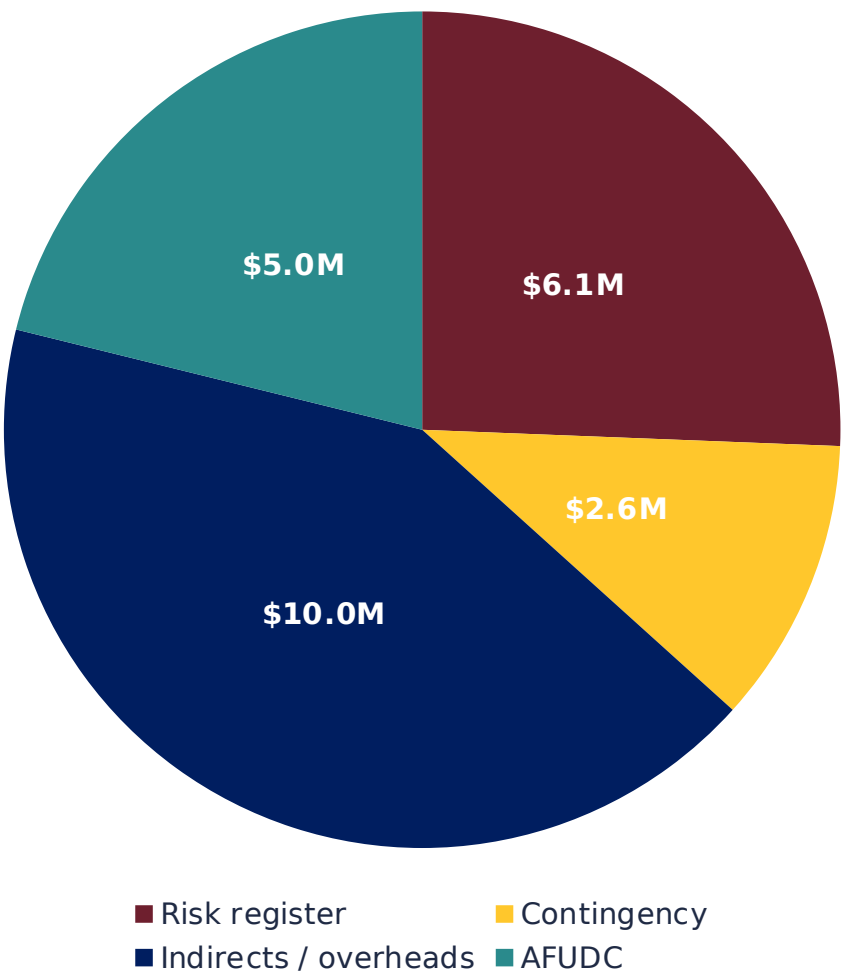


- **Four Siemens SDV7 breaker skids at \$300K each** (relays, PTs, DC battery, cabinet on the skid), disconnects, four metering transclosures, station service, riser poles, 15 kV cable and 1,240 ft of duct bank.
- **Station 496:** 20 SEL relays replacing SEL-551 and REF610 units on four feeders, RTU and RTAC points, transfer trip over dark fiber; \$1.8M of that is testing.
- **Site:** 2.5 acres cleared, 760 ft road, 2,520 CY cut, imported fill, 14 ft retaining wall, ground grid, fence, lighting; Haugland's platform take-off is larger than ours and is unreconciled.
- **D-Line (24211):** 5,465 ft feeder extension, three manholes, 16,395 ft of 3-700 cable; \$4.8M is a 2023 conceptual figure.

**\$18.3M**  
Loaded, in 21334  
32% of the approval level

**\$23.1M**  
Incl. D-Line memo  
37% of the all-in view

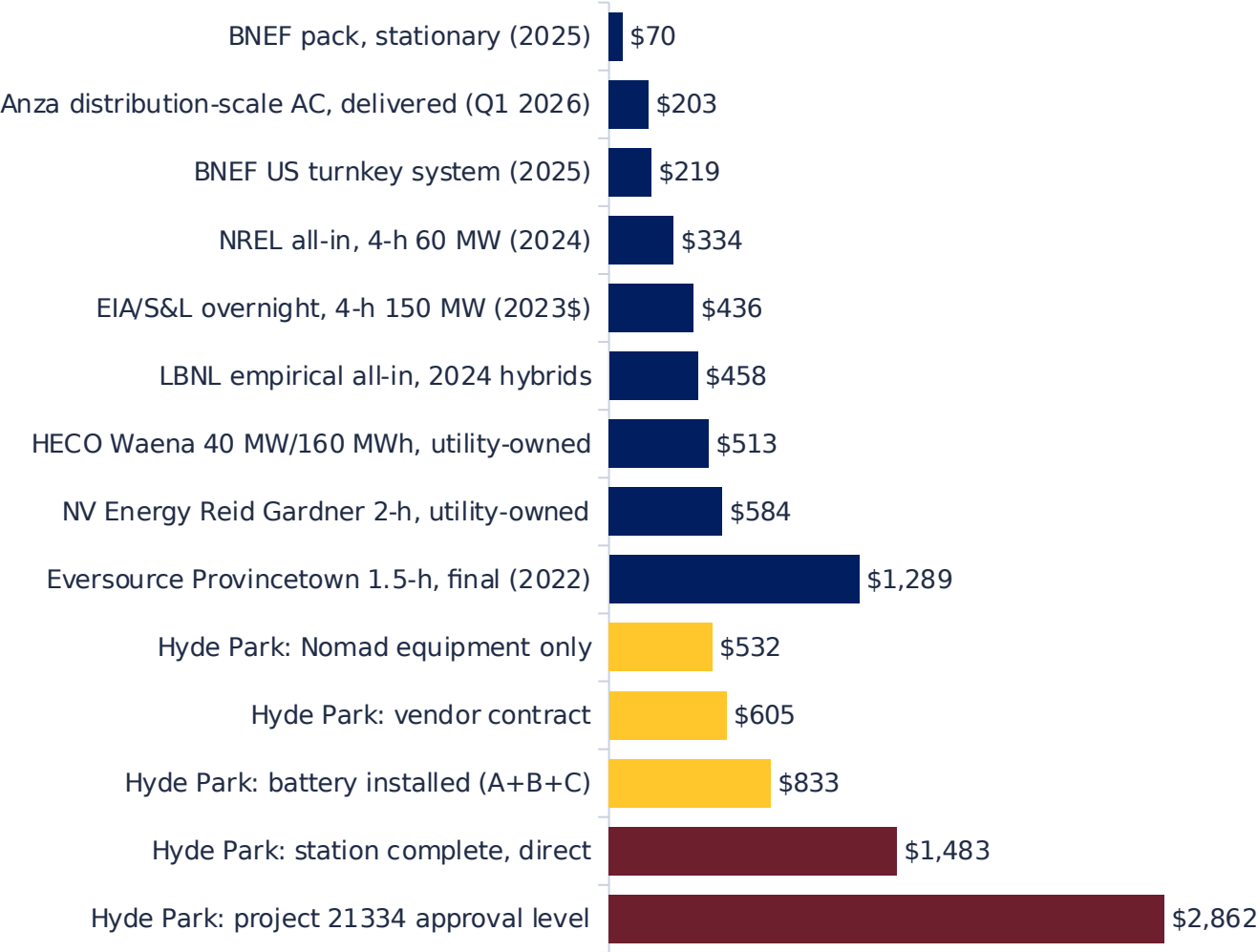
# Risk, contingency, indirects and AFUDC are 41% of the approval level



| Station 360 risk item                            | \$K   |
|--------------------------------------------------|-------|
| Material cost, tariffs                           | 1,000 |
| Design / scope change orders                     | 800   |
| Labor cost, shortage, overtime                   | 800   |
| Unforeseen UG obstructions (landfill)            | 450   |
| Environmental (DEP, landfill closure docs)       | 300   |
| Soil contamination remediation                   | 300   |
| Station and P&C engineering changes              | 600   |
| Weather, siting delay, outreach, BAR deductibles | 1,050 |

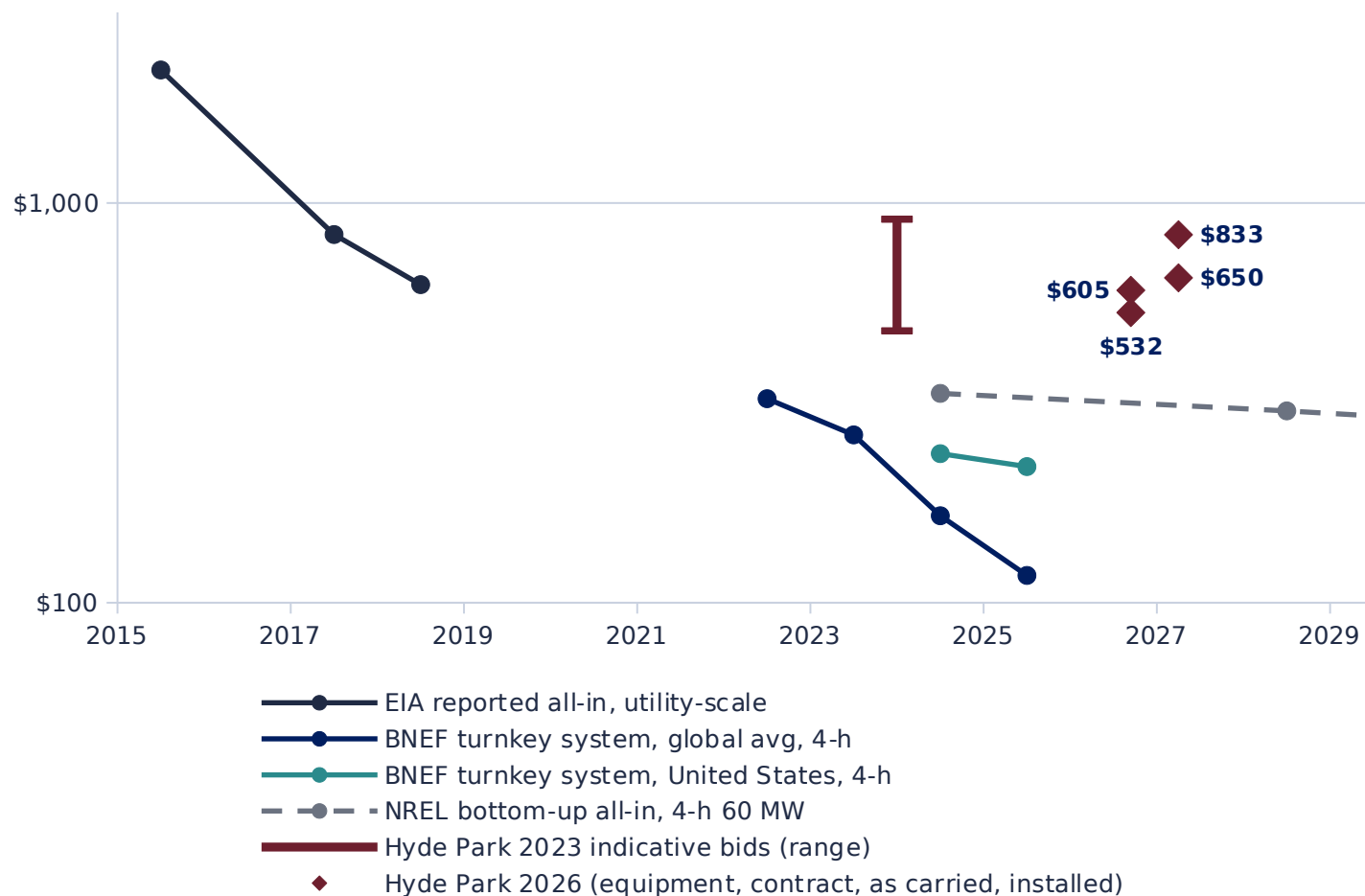
Risk plus contingency is 17.9% of the approval level before loaders, inside the 15-18% lead-review guidance. Indirects at 17.4% and AFUDC at 8.8% follow the EMA station template on a spend curve that runs from July 2021 to December 2028. Haugland's Attachment 4-F carries overlapping exposures (geotech \$0.6M-\$3.5M, fault-study equipment change \$0.25M-\$1.8M) that have not been reconciled with this register.

# The battery is at market; the project is above it for reasons benchmarks exclude



- **Battery at market.** Nomad's \$532/kWh equipment price sits between BNEF's US turnkey (\$219, 4-h, 100 MW class) and the disclosed utility-owned projects (\$513-584). For a 12-unit, 2-h, FEOC-era order in New England that is a fair price.
- **Installed battery (A+B+C) at \$833/kWh** is above every national all-in index because those indices are four-hour and 60-150 MW. NREL's own cost structure (pack \$/kWh x hours + BOS \$/kW) puts a 2-h system at roughly 1.3-1.5x the 4-h \$/kWh.
- **The project at \$2,862/kWh** is not a battery benchmark at all. It carries a switching station, feeder protection at Station 496, a landfill platform, Article 80 siting, indirects and AFUDC. The closest published analogue is Provincetown at \$1,289/kWh (2022 dollars, no separate station, existing site).
- **No published index includes a utility's indirects or AFUDC.** Strip J-M and Hyde Park is \$1,483/kWh direct; strip the station and site layers D and E as well and it is \$1,156/kWh.

# Battery installed-cost curve: the Nomad price sits on it once duration and size are adjusted



■ **The curve is a 4-h, 60-150 MW, real-dollar curve.** EIA's reported all-in fell from \$2,152 (2015) to \$625 (2018); BNEF turnkey from \$324 (2022) to \$117 global and \$219 US (2025); NREL's 4-h bottom-up base is \$334 (2024), \$247 by 2035.

■ **Hyde Park is 2-h, 10 MW, nominal.** NREL's structure puts a 2-h system at 1.3-1.5x the 4-h \$/kWh; the size elasticity in Appendix B adds 1.5x at 20 MWh. Adjusted US turnkey: \$440-\$505. Nomad equipment: \$532.

■ **Install is the layer the curve never shows.** Owner-scope installation, integration and balance of plant (B+C) add \$183/kWh to the \$650 battery line: \$833 installed. BNEF's install sits inside a developer's EPC; ours is Eversource scope at New England rates.

■ **The 2026 turn.** Lithium carbonate roughly doubled year on year by August 2026 and tariffs lifted delivered US prices. The March 2026 fixed price sits below where a 2027 re-procurement would land.

Plotted as published: EIA-860 nominal; BNEF real survey-year; NREL 2024\$ (2028 interpolated between the 2024 base and the 2035 mid case). 2023 bar = five indicative bids, \$479-\$911/kWh, differing scopes. Retrieved from search excerpts; verify before external use.

# Escalation is already in the number: \$2.9M at template rates, \$1.1M of it on a fixed price

## Where the escalation sits (Rev 3, included in the directs)

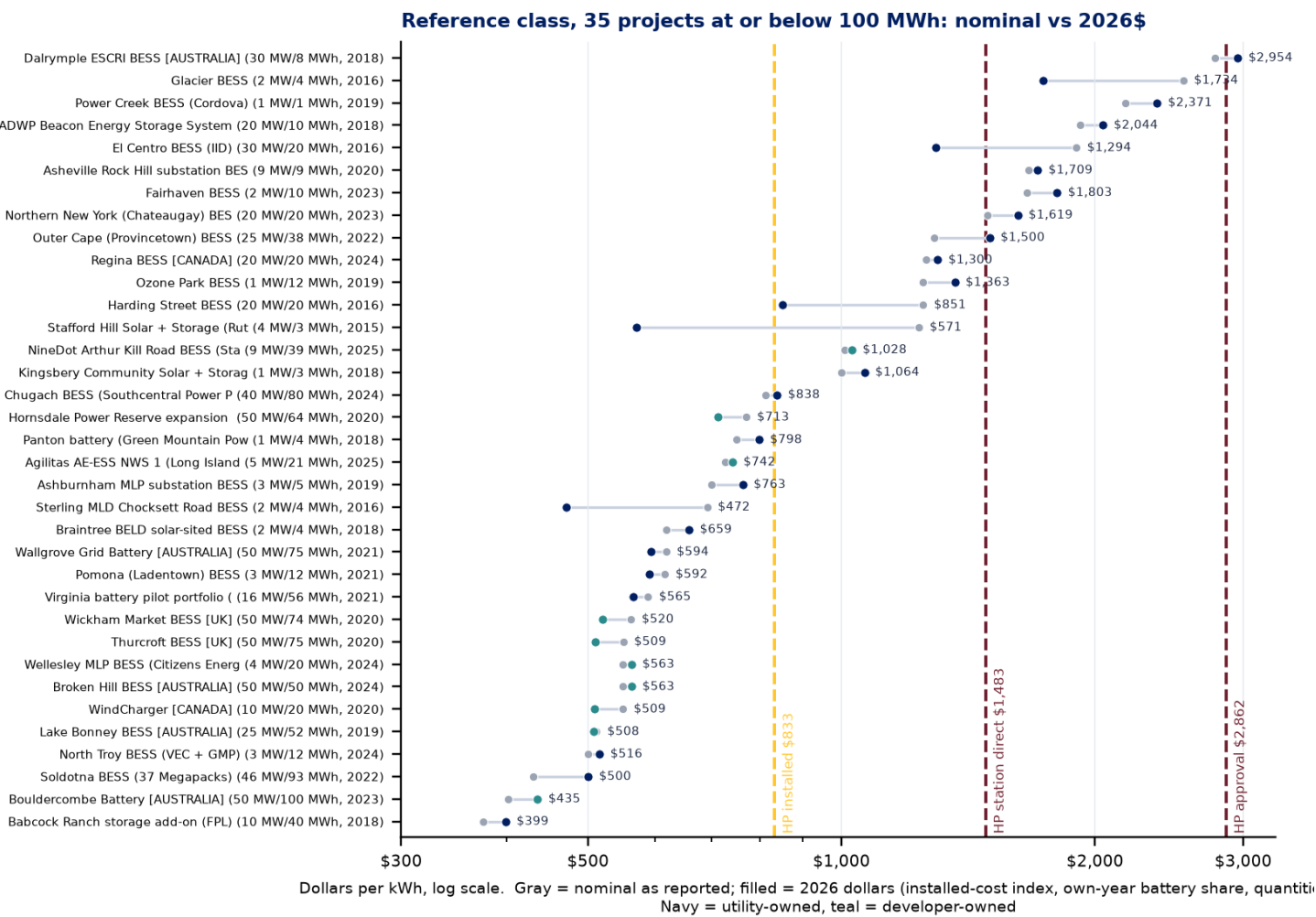
| Item                                                                                   | \$       | \$/kWh |
|----------------------------------------------------------------------------------------|----------|--------|
| Escalation in the approval level (Sta 360 \$2,622K + Sta 496 \$326K)                   | \$2,948K | \$147  |
| on the Nomad balance, which Exhibit C fixes until Final Acceptance                     | \$1,130K | \$57   |
| on station, site, feeder protection, testing and owner scope                           | \$1,818K | \$91   |
| Spend curve it is applied to: actuals \$5.2M, 2026 \$13.5M, 2027 \$12.5M, 2028 \$26.1M |          |        |

## Same comparison, same dollars (per kWh)

| Boundary                  | Nominal | Ex-escalation | Benchmark brought to 2028 at 3%/yr                 |
|---------------------------|---------|---------------|----------------------------------------------------|
| Vendor contract           | \$605   | \$605 (fixed) | BNEF US turnkey \$219 (2025) = \$239               |
| Battery as carried (A)    | \$650   | \$593         | NREL 4-h \$334 (2024\$) = \$376                    |
| Battery installed (A+B+C) | \$833   | ~\$756        | NREL adjusted to 2-h, 10 MW: \$750-\$870           |
| Station complete, direct  | \$1,483 | \$1,352       | EIA/S&L \$436 (2023\$) = \$505; LBNL \$458 = \$516 |

- **Rates.** ESF template escalation, about 3% a year, on the 2026-2028 spend curve. \$2.9M is 5.1% of the approval level and 7.4% of directs. Benchmarks are 2023-2025 dollars; Rev 3 is 2026-2028 nominal, so one side must move before they are compared.
- **Decision pending.** \$1.13M sits on a price the contract fixes. Remove it or move it to the risk register with a stated rationale (open item, slide 37).
- **Brought to the same year, the installed battery is on the index.** NREL's \$334 is \$376 in 2028 dollars and \$750-\$870 after the 2-h and 10 MW adjustments; Hyde Park installed is \$833.
- **What 3% does not cover.** Lithium carbonate and tariffs moved faster than 3% in 2026. The battery is fixed by contract, so the exposure is the \$3.7M of non-battery material and the labor lines. The \$1.0M tariff item in the risk register is sized on total material, battery included.
- **Schedule.** Twelve months of slip adds about \$1.2M of escalation on the \$38.6M scheduled for 2027-2028, before AFUDC, PM and indirects (about \$5M in total).
- **Ex-escalation values** strip the station's \$131/kWh from the direct boundaries; the B+C share (~\$20/kWh) is allocated pro rata and is approximate.

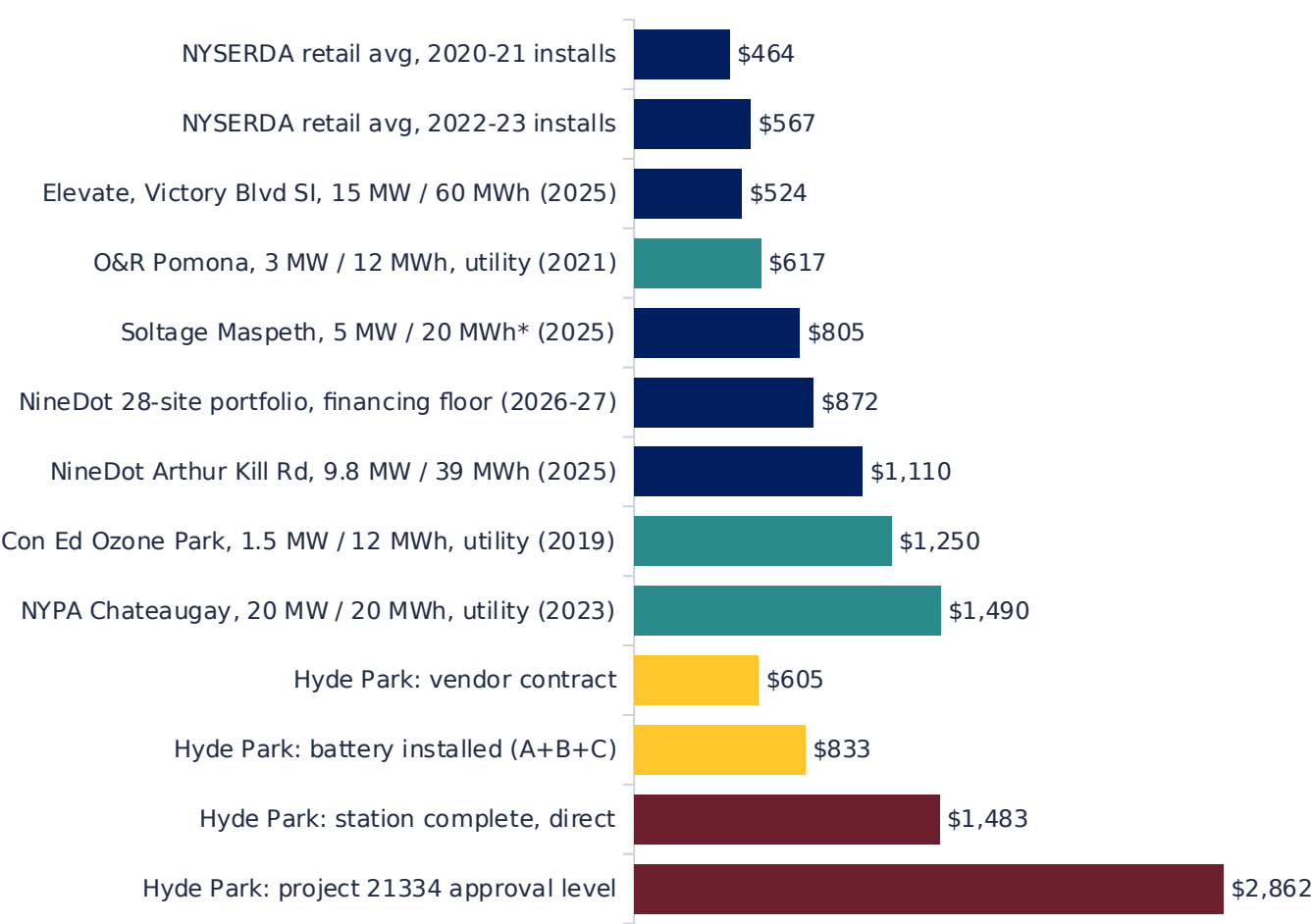
# Own-year battery shares: the re-based class median barely moves and Hyde Park's station cost stays in the top fifth



| Battery index (own-year share) | Median \$/kWh | HP installed | HP station |
|--------------------------------|---------------|--------------|------------|
| Nominal, as published          | \$750         | 57%          | 77%        |
| Installed all-in (EIA, LBNL)   | \$742         | 57%          | 77%        |
| BNEF pack price                | \$743         | 60%          | 83%        |
| BNEF turnkey system            | \$686         | 57%          | 83%        |
| NREL bottom-up base year       | \$742         | 54%          | 80%        |
| Lazard LCOS midpoint           | \$892         | 40%          | 66%        |
| Construction index only        | \$1,052       | 40%          | 63%        |

- **The weights move with the prices.** Quantities are held fixed, not today's dollar shares: a 2016 project was about 73% battery and comes to 2026 at 0.68x; a 2018 project was 56% battery, 1.06x; 2021 and later about 1.0x. The 5 Oct review challenge stands: the earlier version held a 40% share fixed and made every old project dearer (median \$838).
- **Widening the class to 35 projects, 10 developer-owned,** puts Hyde Park's station-direct cost at the 77% percentile nominal and 77% re-based; the developer sites carry no utility station, no landfill and no siting case.
- **The battery curve you pick moves the median by 8%;** slides 14 to 18 show the curves and what moves it more.

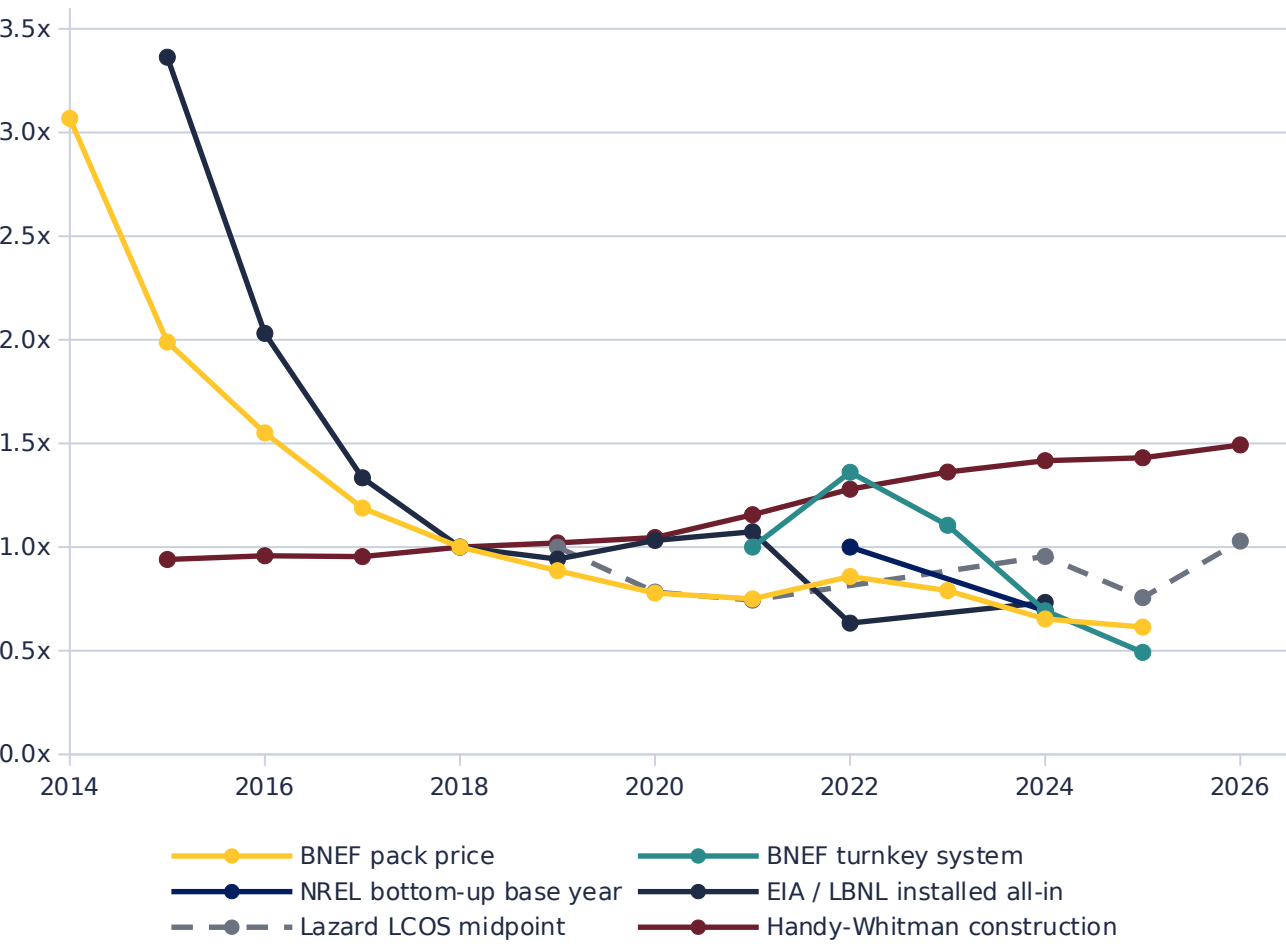
# New York is building the same thing: distribution-connected, 2025, \$524 to \$1,110 per kWh before any utility station



| Per kW of power                        | \$/kW   |
|----------------------------------------|---------|
| NineDot Arthur Kill Rd (NYCIDA budget) | \$4,427 |
| NineDot portfolio financing (floor)    | \$3,476 |
| Soltage Maspeth (NYCIDA)               | \$3,221 |
| Elevate Victory Blvd (NYCIDA)          | \$2,096 |
| Hyde Park station complete, direct     | \$2,967 |
| Hyde Park approval level               | \$5,725 |

- **NineDot:** 11 projects at 6 NYC sites, 39 MW operating (Sep 2026); 28 more financed at \$431M for 494 MWh. Thirteen NYC developer sites with NYCIDA-filed costs are in the dataset (NineDot, Convergent, Soltage, Agilitas, Greenbacker, Elevate): \$410 to \$1,110/kWh, median about \$830. Developer-owned 4-h sites on leased lots, VDER revenue, NYSERDA incentive, ITC; no utility station, no landfill, no siting case.
- **Their own filed budget for Arthur Kill Road is \$1,110/kWh and \$4,427/kW**, inside Hyde Park's installed-to-station band on \$/kWh at twice the duration.
- **Interconnection is the cost no index carries:** Con Edison's two-part test added about \$21M of upgrades per project on 34 projects (161 MW), per the NY-BEST and NYSEIA filing of 11 Mar 2026. Hyde Park's Station 496 scope is \$7.3M.
- **\*Soltage duration assumed 4-h;** incentives and ITC not netted from any row.

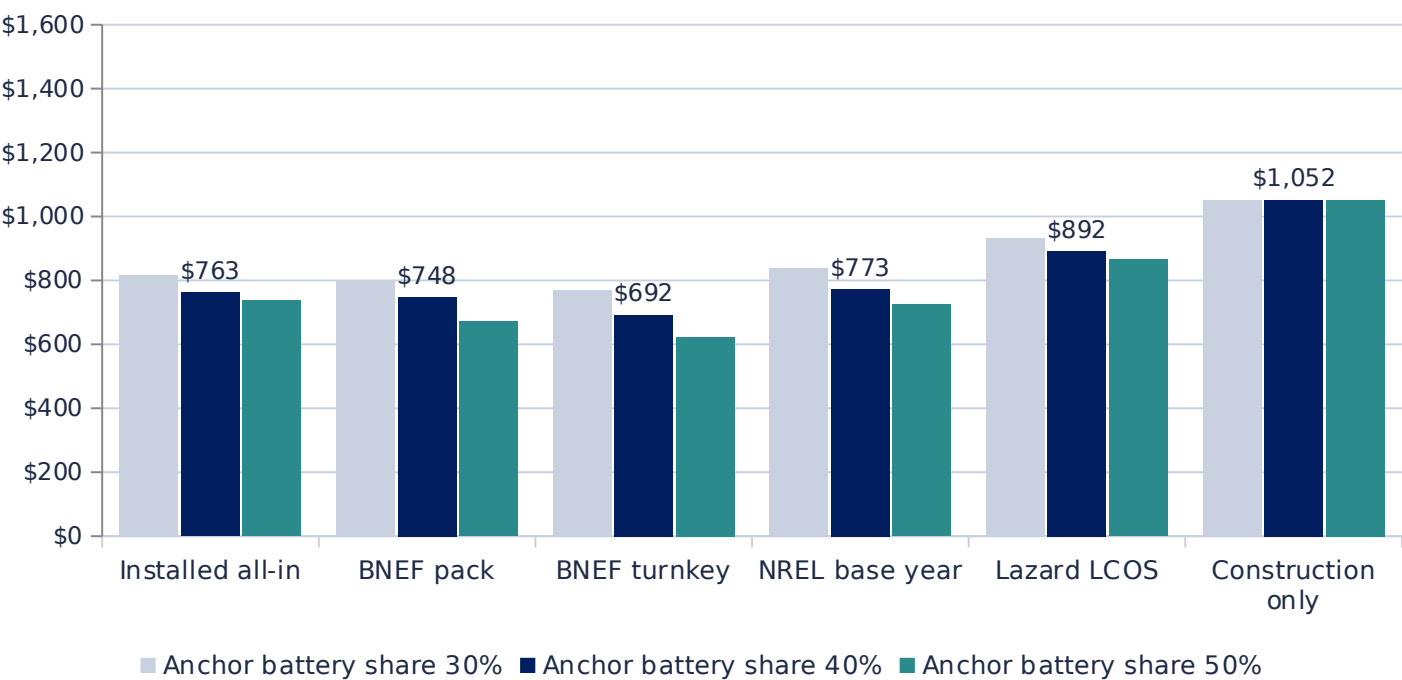
# Six published curves, indexed to 2018: batteries fell 1.4 to 2x, the rest of a project rose half again



| Curve                       | What it measures                                                                  | 2018 to latest | Use in re-basing              |
|-----------------------------|-----------------------------------------------------------------------------------|----------------|-------------------------------|
| BNEF pack price             | Cells and pack, ex-works; real \$; annual since 2010                              | 0.61x          | Battery share only            |
| BNEF turnkey system         | Racks, PCS, BOS, install; excl. grid connection, land, owner costs                | 0.49x          | Battery share; short history  |
| NREL bottom-up base year    | Modeled 4-h 60 MW all-in incl. EPC, developer overhead, interconnection allowance | 0.69x          | Battery share; two points     |
| EIA / LBNL installed all-in | Reported capitalized cost of installed systems (EIA-860 2015-18; LBNL 2024)       | 0.73x          | Primary: closest to a project |
| Lazard LCOS                 | Levelized \$/MWh incl. charging, O&M, augmentation, financing, ITC                | 1.03x          | Not a capital-cost deflator   |
| Handy-Whitman construction  | Utility plant construction cost index, North Atlantic (via PJM)                   | 1.49x          | Non-battery share (60%)       |

■ **Four curves describe the battery, one the rest of the project, one neither.** A re-basing picks one from each of the first two groups and says which. LCOS is cited most and fits least.

# The battery curve you choose moves the median 8%; the battery share and the construction index move it more



Nominal median \$750. Hyde Park station complete, direct, is \$1,483/kWh; its percentile under each index is at right. Deflating whole projects at the battery rate gives \$390 to \$549.

Indices as published (BNEF pack and ESS surveys 2014-2025; EIA-860 2015-18 and LBNL 2024; NREL 2022 and 2024 base years; Lazard LCOS v5.0-2026 midpoints; Handy-Whitman via PJM), linearly interpolated between published points and held flat outside them. Retrieved from search excerpts; verify before external use.

| Index             | 2018 to 2026 | Factor to 2026: 2018 / 2016 project | HP station pct |
|-------------------|--------------|-------------------------------------|----------------|
| Installed all-in  | 0.73x        | 1.06x / 0.68x                       | 77%            |
| BNEF pack         | 0.61x        | 0.97x / 0.74x                       | 83%            |
| BNEF turnkey      | 0.49x        | 0.92x / 0.93x                       | 83%            |
| NREL base year    | 0.69x        | 1.03x / 1.05x                       | 80%            |
| Lazard LCOS       | 1.03x        | 1.26x / 1.29x                       | 66%            |
| Construction only | 1.49x        | 1.49x / 1.56x                       | 63%            |

- **Choice of battery curve:** medians \$686 to \$743 across the four battery series (LCOS \$892). The curves disagree about packs, not about projects.
- **Anchor battery share:** 30% to 50% (2024 basis) moves the median \$79 to \$146 depending on the curve; the share is a lever of the same order as the curve.
- **Construction index:** Handy-Whitman North Atlantic rose 49% from 2018 to 2026 (10% a year in 2021-22); a flat 3.5% proxy gives 32%, and a median \$742 instead of \$742.
- **LCOS** is a levelized \$/MWh metric, flat since 2019; using it re-bases nothing on the battery side.

## Whichever curve you cite, Hyde Park's station cost stays in the top quarter and the approval level stays above the class

| Curve used to re-base       | Class median | HP installed pct | HP station pct | Utility subclass median | HP station pct (utility) | Whole cost deflated at battery rate: median / HP station pct |
|-----------------------------|--------------|------------------|----------------|-------------------------|--------------------------|--------------------------------------------------------------|
| Nominal, as published       | \$750        | 57%              | 77%            | \$1,235                 | 68%                      | n/a                                                          |
| EIA / LBNL installed all-in | \$742        | 57%              | 77%            | \$851                   | 68%                      | \$549 / 86%                                                  |
| BNEF pack price             | \$743        | 60%              | 83%            | \$929                   | 76%                      | \$517 / 94%                                                  |
| BNEF turnkey system         | \$686        | 57%              | 83%            | \$915                   | 76%                      | \$390 / 100%                                                 |
| NREL bottom-up base year    | \$742        | 54%              | 80%            | \$1,189                 | 72%                      | \$550 / 91%                                                  |
| Lazard LCOS midpoint        | \$892        | 40%              | 66%            | \$1,339                 | 52%                      | \$875 / 74%                                                  |
| Construction index only     | \$1,052      | 40%              | 63%            | \$1,493                 | 48%                      | n/a                                                          |

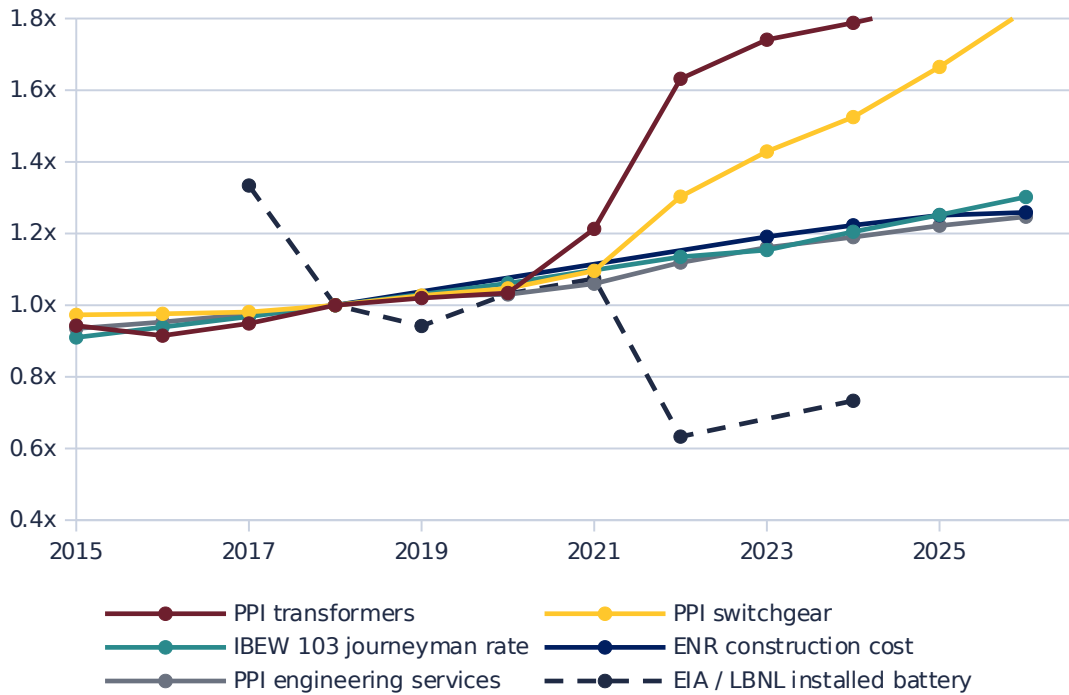
■ **Reading across:** the four battery curves place the class median within \$56/kWh of each other and Hyde Park's station cost between the 77% and 83% percentiles. LCOS and a construction-only index raise the class because they carry no battery deflation.

■ **The last column is the challenge as it was put to us:** deflate every comparable at the battery rate. It moves the median to \$390 to \$550 and Hyde Park to the top of the class, by treating switchgear, civil works and interconnection as lithium. It is the vendor framing applied backward in time.

■ **Which to cite:** the EIA / LBNL installed series to a regulator (reported costs of real installations); the BNEF pack series to a vendor (it is their number); neither LCOS nor a flat proxy to anyone for re-basing capital cost.

# One escalator per layer, quantities fixed: a 2018 project like Hyde Park costs +4% in 2026 dollars, a 2016 one 32% less

| Layer (loaded share)       | Share | Escalator                                             | 2018 to 2026 |
|----------------------------|-------|-------------------------------------------------------|--------------|
| A. Battery contract        | 38%   | EIA / LBNL installed battery                          | 0.73x        |
| B. Battery install         | 5%    | IBEW 103 journeyman rate                              | 1.30x        |
| C. GSUs, aux power         | 6%    | PPI transformers                                      | 1.91x        |
| D. Switching station       | 11%   | PPI switchgear                                        | 1.82x        |
| E. Site and civil          | 8%    | ENR construction cost                                 | 1.26x        |
| F. Feeder protection (496) | 13%   | IBEW 103 journeyman rate                              | 1.30x        |
| G. Test and commission     | 4%    | IBEW 103 journeyman rate                              | 1.30x        |
| H. Owner soft costs        | 13%   | PPI engineering services                              | 1.25x        |
| I. Taxes, insurance        | 2%    | Handy-Whitman                                         | 1.49x        |
| Composite, Hyde Park scope | 100%  | fixed quantities (1 / weighted sum of inverse ratios) | 1.04x        |



Each series divided by its 2018 value. BLS PPI series are exact to 2023 (FRED mirror), excerpt anchors after; ENR and IBEW from published tables.

- **Five non-battery escalators sit between 1.25x and 1.91x since 2018; the battery sits at 0.73x.** With Hyde Park's physical scope held fixed, a 2018 build comes to 2026 at 1.04x (two-index model 1.06x) and a 2016 build at 0.68x, because the battery was 72% of such a project in 2016 and is 38% now. Transformers (1.91x) and switchgear (1.82x) are the fastest risers.
- **The review challenge was right about 2015 to 2017, and the earlier version of this slide was wrong.** Holding today's 38% battery share fixed made a 2016 project 1.08x; letting the share be what it was then makes it 0.68x. From 2018 on the battery decline and the construction rise roughly cancel. Deflating a whole project at the pack rate is still wrong: it gives 0.36x for 2016.

# Three defensible methods, one answer: class median \$742 to \$746, Hyde Park's station cost at the 77% percentile

| Method                                        | What it does                                       | Class median | HP installed pct | HP station pct | Utility median | HP station pct (utility) |
|-----------------------------------------------|----------------------------------------------------|--------------|------------------|----------------|----------------|--------------------------|
| As published                                  | no adjustment                                      | \$750        | 57%              | 77%            | \$1,235        | 68%                      |
| Whole cost at the battery rate                | vendor framing applied backward                    | \$549        | 69%              | 86%            | \$685          | 80%                      |
| Two indices, own-year battery share           | anchor 40% utility, 55% developer; slides 12 to 16 | \$742        | 57%              | 77%            | \$851          | 68%                      |
| One index per layer, developer weights        | NREL/PNNL composition                              | \$744        | 60%              | 77%            | \$854          | 68%                      |
| One index per layer, Hyde Park weights        | Rev 3 loaded layers                                | \$746        | 57%              | 77%            | \$853          | 68%                      |
| Layer method, BNEF pack for the battery       | sensitivity                                        | \$680        | 63%              | 83%            | \$832          | 76%                      |
| Layer method, BNEF turnkey for the battery    | sensitivity                                        | \$585        | 63%              | 89%            | \$770          | 84%                      |
| Earlier version: today's 40% share held fixed | superseded 5 Oct 2026                              | \$838        | 49%              | 71%            | \$1,285        | 60%                      |

**How to scale any published BESS cost to 2026 (the workbook tab does it)**

- **1. Fix the dollar year and the scope boundary.** Contract, installed, station or all-in; real or nominal; which year's survey.
- **2. Fix the battery share in today's prices, then back-cast it.** 40% for an all-in utility project, 55% for a developer site; the two indices tell you what that share was in the project's year (about three quarters in 2016).
- **3. Apply a battery index to the battery share and a construction or layer index to the rest, holding quantities fixed.** Never one index to the whole cost, and never today's shares on yesterday's prices.

- **Method spread:** \$742 to \$746 across the three defensible methods; \$549 for the naive one.
- **Hyde Park:** installed 57% to 60% percentile, station 77% to 77%, under every defensible method.

## Six things about Hyde Park that a 100 MW greenfield benchmark never has to pay for

**Size.** 10 MW / 20 MWh. A switching station, a siting case and a commissioning team cost about the same at 10 MW as at 100 MW; per kWh they are ten times heavier.

**Duration.** Two hours. Fixed \$/kW items (PCS, breakers, GSUs, interconnection) are spread over half the MWh of a four-hour benchmark.

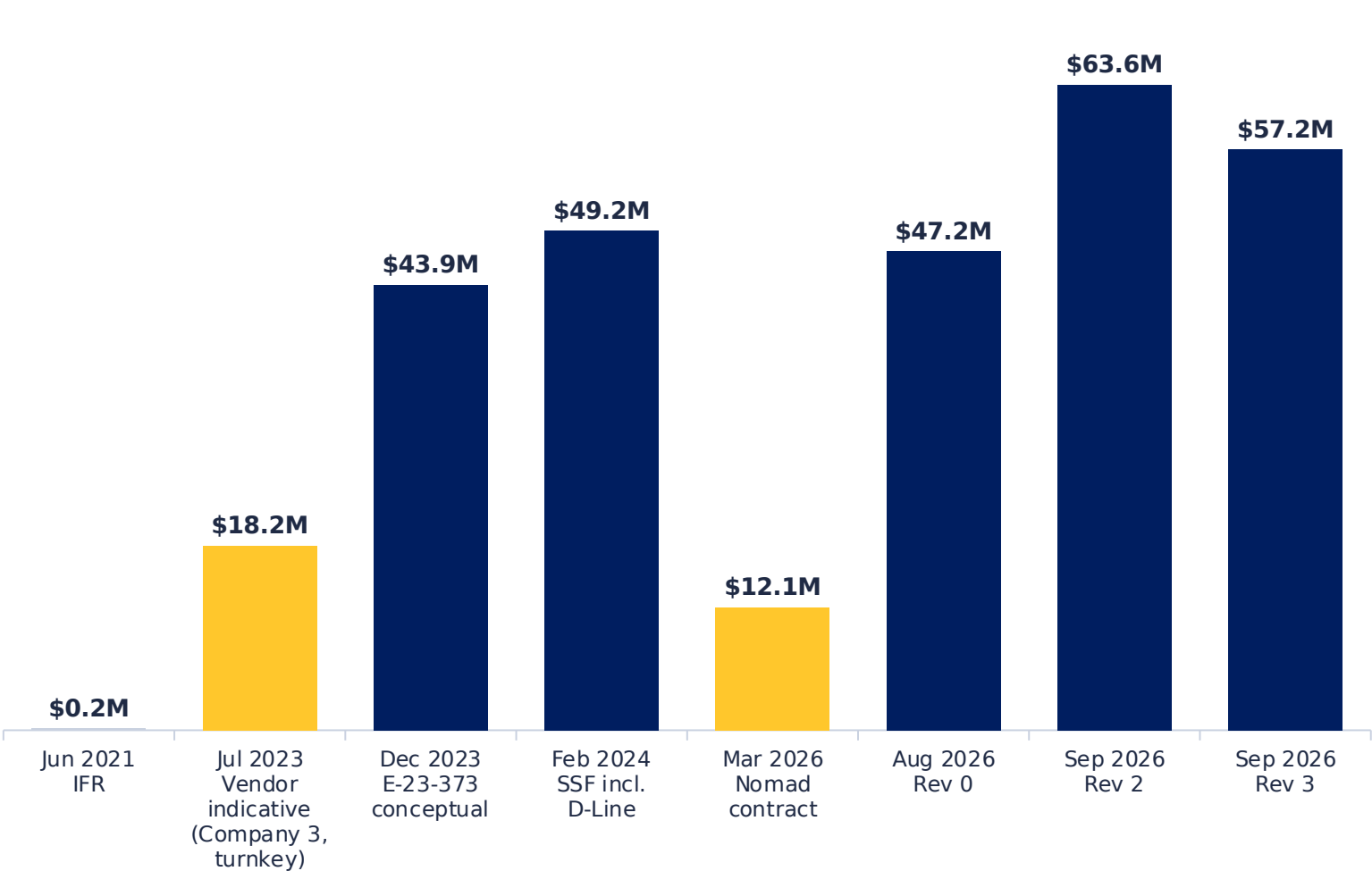
**Site.** A closed construction-debris landfill in Boston: DEP post-closure use permit, cap protection, 36 borings, imported platform fill, 14 ft retaining wall, unknown obstructions (\$450K risk).

**Interconnection.** Four feeder tie-ins at 13.8 kV, each with its own breaker skid, plus relay replacement and transfer trip at Station 496 (\$7.3M) and a 5,465 ft circuit extension (24211).

**Siting and outreach.** Article 80 Small Project Review, Zoning Board of Appeal, outside counsel, an EJ community, a neighborhood concerned about open land and views: \$1.5M plus \$0.5M in the risk register.

**Utility accounting.** Corporate indirects (\$10.0M) and AFUDC over a seven-year development window (\$5.0M) are real costs to customers that a developer's turnkey price never shows.

# Five years, five in-service dates, and an estimate that moved from \$44M to \$57M

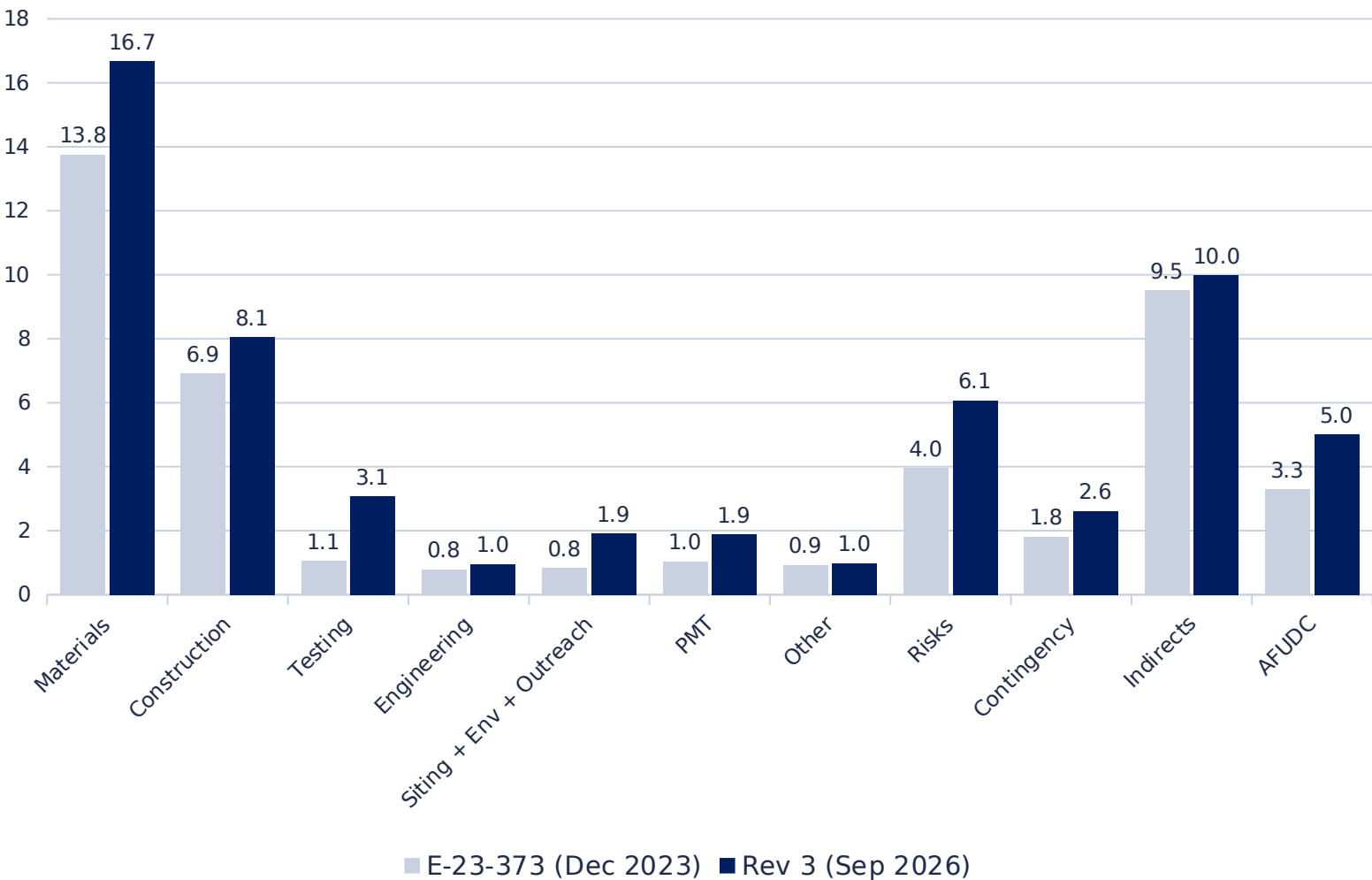


## In-service date drift

| As of    | Planned ISD |
|----------|-------------|
| Jun 2021 | Q3 2025     |
| Dec 2023 | Apr 2027    |
| Feb 2024 | Sep 2027    |
| Jan 2026 | Aug 2028    |
| Jun 2026 | Dec 2028    |

Gold bars are vendor prices for the battery scope only. Navy bars are full project estimates. The battery price fell from \$18.2M turnkey-indicative (2023, incl. civil) to \$12.1M contracted (2026); the project rose \$13.3M in the same window. The 2023 conceptual carried a placeholder that was never re-based until August 2026.

# What changed between the December 2023 conceptual and Rev 3: \$43.9M to \$57.2M

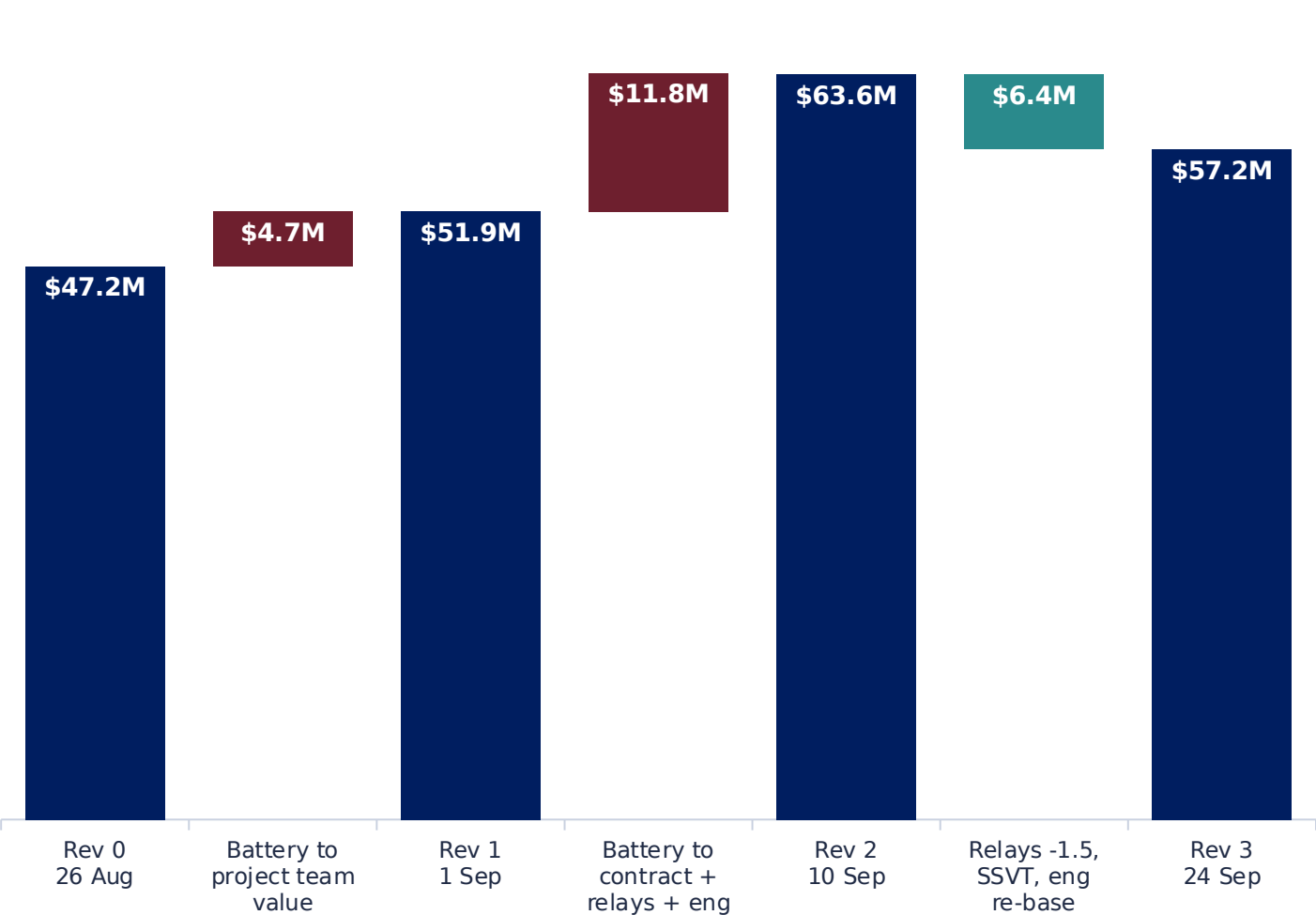


## Largest movements

| Category                | Delta   |
|-------------------------|---------|
| Materials               | +\$2.9M |
| Risks                   | +\$2.1M |
| Testing                 | +\$2.0M |
| AFUDC                   | +\$1.7M |
| Construction            | +\$1.1M |
| Siting + Env + Outreach | +\$1.1M |
| PMT                     | +\$0.9M |

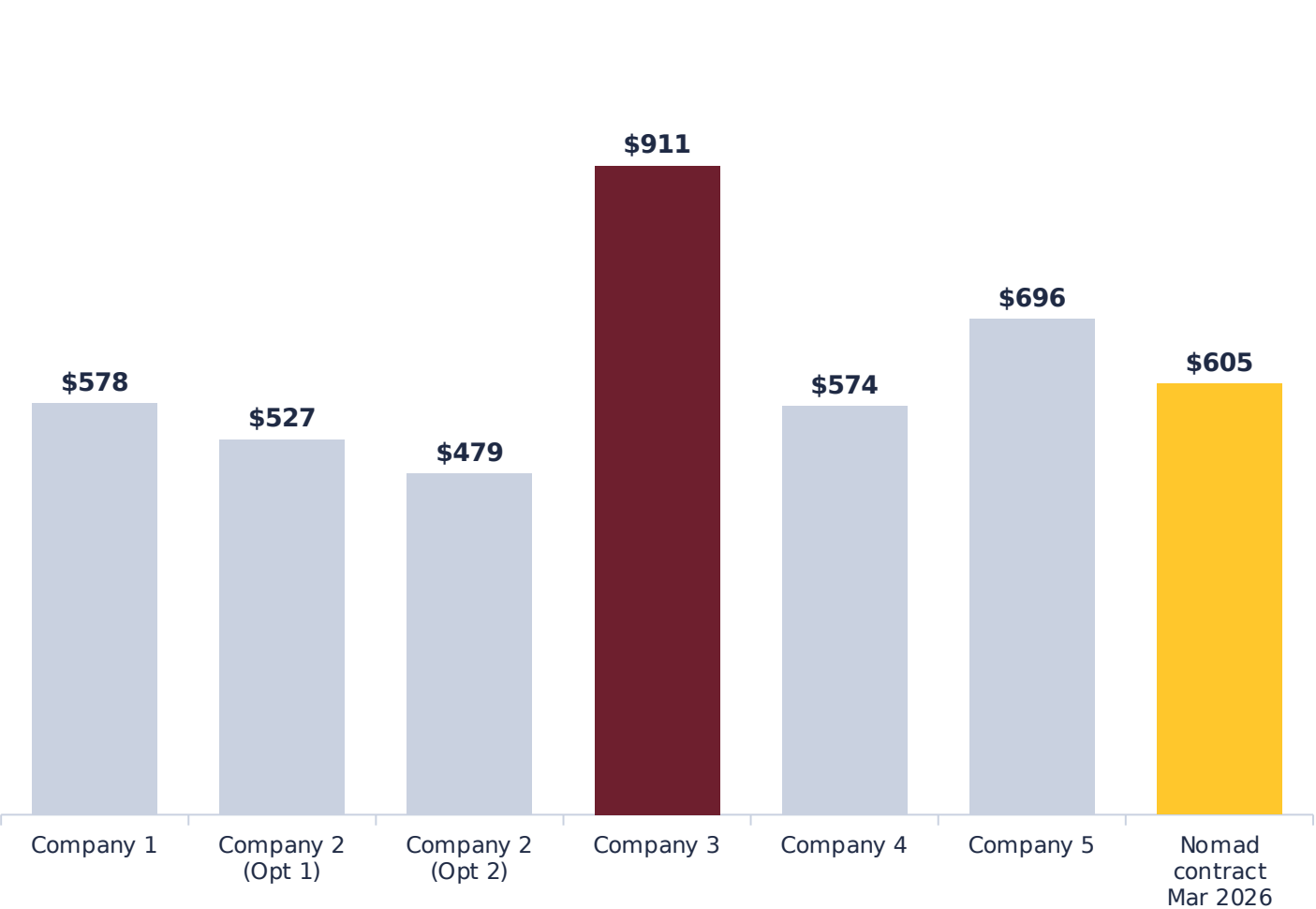
Materials: contracted battery replaces 2023 indicative pricing, plus breaker skids and GSUs. Testing and siting: Station 496 scope and Article 80 were not in E-23-373. AFUDC: a 20-month later ISD on a larger base.

# Four revisions in five weeks: how the September estimate settled at \$57.2M



- **Rev 0 carried the battery at about \$6M** from the 2023 indicative sheet. The PM: 'we know the BESS system itself is 12.'
- **Rev 2 reset it to the cover agreement, escalated to \$13.1M**, and carried \$1.5M of 'relays for batteries' from a 2023 bidder that priced its own switchgear.
- **Rev 3 removed the relays** (P&C Engineering: included in the \$300K Siemens skid), cut three SSVTs to one, and re-based engineering to \$501K to go plus actuals.
- **Net:** +\$10.0M from Rev 0, of which \$7.0M is the battery line catching up to the contract.

# The battery: a \$9.6M to \$18.2M spread in 2023, a \$12.1M fixed price in 2026



- **July 2023, five bidders, 10 MW / 20 MWh:** \$9.6M to \$18.2M. Every bidder excluded interconnection, permits beyond local, below-grade risk, sound walls or spares.
- **Company 3 (\$18.2M, burgundy)** became the E-23-373 basis because it alone priced turnkey civil (\$4.56M) and installation.
- **March 2026 contract, \$12.1M (gold):** supply only, with supervision. Civil, install, GSUs, station and interconnection moved to Eversource scope.
- **Equipment-only inside the contract:** \$10.64M, \$532/kWh, in line with published 2025-26 turnkey system indices (slides 9-10).
- **Rev 3 adds \$1.13M of escalation** to a price the contract fixes until Final Acceptance.

## Three assumptions that, if wrong, move the number more than the reserve

### 1. Construction may be low, not high.

Rev 3 layers B+D+E are \$8.3M direct. Haugland's Class 3 base for overlapping scope is \$19.2M with a \$2.9M deduct ceiling. Nobody has mapped one to the other.

*Settles it: a signed HEG-to-Rev 3 scope map before the PAF is drafted.*

### 2. Siting is binary and carries no float.

ZBA determination, construction start and container delivery all land in Q1 2028 for a Dec 2028 ISD. Denial means an EFSB exemption at 18-24 months. The estimate carries \$250K for siting delay.

*Settles it: Article 80 filing date, ZBA calendar, P6 float on the siting chain.*

### 3. The number is Class 4 wearing Class 3 clothes.

30% engineering, unit-library construction with no bid, install labor as a percentage of a 2023 total, a 2.5 ft platform placeholder. EPAC will hear \$57.2M as a planning number.

*Settles it: geotech, HMA, HEG map, crew build-up; then reclassify with a -15/+25% range.*

# Stress tests on our own numbers, and what history says about small utility batteries

| Scenario                                                  | Loaded cost         | Date      | Covered by \$7.6M reserve? |
|-----------------------------------------------------------|---------------------|-----------|----------------------------|
| Construction unit costs +25% on B+D+E+G                   | +\$4.0M             | 0         | Yes, alone                 |
| HEG reconciliation at midpoint of the gap (+ \$5M direct) | +\$8.4M             | 0         | No                         |
| Schedule +12 months (siting or long-lead)                 | ~+\$5M              | +12 mo    | Partly                     |
| EFSB zoning exemption path                                | ~+\$8M to + \$10M   | +18-24 mo | No                         |
| Contractor bids +20% over estimate labor                  | +\$2.1M             | 0         | Yes, alone                 |
| Combination: HEG midpoint + 12-month slip                 | ~+\$13M (to ~\$70M) | +12 mo    | No                         |
| Battery change orders at the LD cap (10%)                 | +\$1.2M             | 0         | Yes                        |

Untestable from the package: tariff or change-in-law pass-through on the Nomad price (Exhibit C terms), the AFUDC rate (assumed ~7% on CWIP), Station 496 outage premium (1.35 to 1.70 factor on affected hours), FEOC / ITC effect on net cost.

## Base rate: approved to actual

| Project                           | Uplift      |
|-----------------------------------|-------------|
| BGE Fairhaven 2.5 MW/9.7 MWh      | +64%        |
| HECO Waena 40 MW/160 MWh          | +37%        |
| SaskPower Regina 20 MW            | +31%        |
| Eversource Provincetown           | +9% to +40% |
| Hyde Park, Dec 2023 to Sep 2026   | +30%        |
| Hyde Park, Rev 0 to Rev 3 (5 wks) | +21%        |

Rev 3 carries 17.9% risk plus contingency. The exception case for beating the base rate rests on three anchors: a fixed battery price, a Siemens price on the skids, and Haugland engaged pre-construction. Together they cover about a third of the cost. The other two thirds have no such anchor.

# Pre-mortem: June 2031, \$74M, fifteen months late. What happened?

*The ZBA conditioned approval on a smaller footprint and new screening; the civil package went back through Haugland in the winter of 2028. Spring 2027 borings found debris deeper than the 2014 closure set showed; the platform went from 2.5 ft of fill to piles, and the reconciliation that had never been done was done as a change order. Nomad's twelve containers arrived on time in January 2028 and sat in a rented yard for nine months. Station 496 cutovers needed night outages priced at straight time. AFUDC alone added \$4M. The PAF had said Conceptual, -25/+50%. Everyone remembered \$57M.*

## Early-warning indicators

| Indicator                                                          | Caught today?                                                         |
|--------------------------------------------------------------------|-----------------------------------------------------------------------|
| Geotech report not issued by IFC-civil design freeze (mid-2027)    | No: no P6 gate ties IFC to it                                         |
| ZBA hearing date not set within 90 days of Article 80 filing       | No: siting carries hours, not milestones                              |
| HEG reconciliation not closed before the PAF is drafted (Nov 2026) | Only if the CoE makes it a precondition                               |
| Nomad final-drawings milestone (13 Jul 2026) slipped               | Partly: payments tracked, not downstream effect on GSU spec and civil |
| Station 496 outage plan not issued before construction pricing     | No                                                                    |

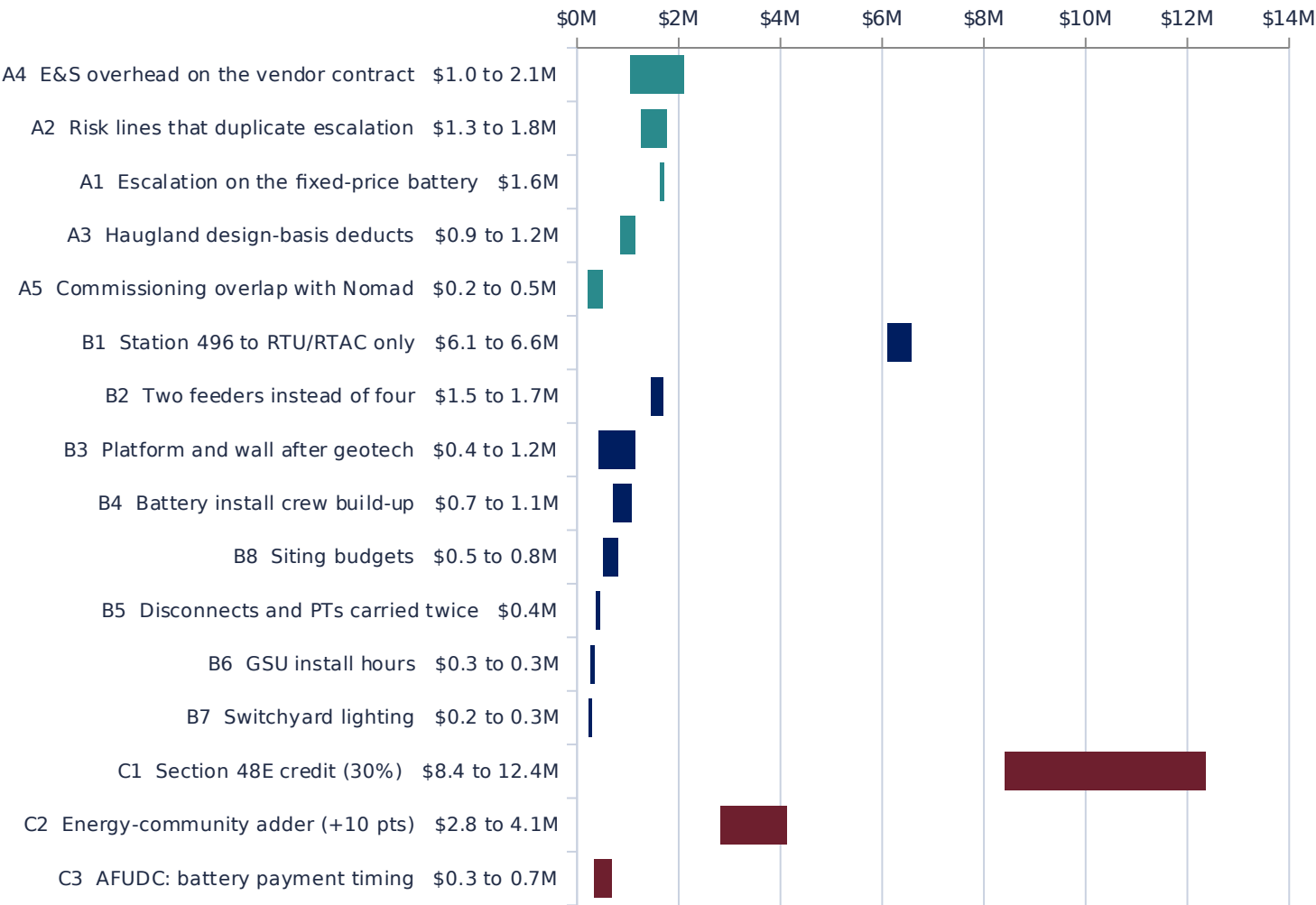
# 16 levers in three tiers: \$5.0 to 7.1M in the estimate, \$10.1 to 12.3M in scope decisions, and a tax credit larger than both

| #  | Lever                                                                                                          | Layer     | Direct         | Loaded*        | Conf.  | What must close                                                                                                                                                                                                                                                          |
|----|----------------------------------------------------------------------------------------------------------------|-----------|----------------|----------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A1 | Remove escalation carried on the fixed-price Nomad balance                                                     | A         | \$1.1M         | \$1.6M         | High   | Exhibit C read for tariff and change-in-law pass-through; decision minuted                                                                                                                                                                                               |
| A2 | Risk register: labor-cost and material-cost lines overlap the escalation already in the directs                | J         | \$1.3 to 1.8M  | \$1.3 to 1.8M  | Medium | Re-word the two lines as labor-shortage and tariff risks and size them on exposure, not on the whole material and labor base                                                                                                                                             |
| A3 | Confirm the four written design-basis items Haugland priced as deducts                                         | D/E       | \$0.6 to 0.8M  | \$0.9 to 1.2M  | Medium | Retaining wall, basin, duct bank and GSU count confirmed in writing; HEG map signed first                                                                                                                                                                                |
| A4 | Engineering & supervision overhead basis applied to the vendor turnkey contract                                | L         | \$1.0 to 2.1M  | \$1.0 to 2.1M  | Low    | Ruling on whether the E&S station basis applies to vendor-furnished, vendor-commissioned equipment                                                                                                                                                                       |
| A5 | Commissioning hours carried from E-23-373 overlap Nomad's commissioning support                                | G         | \$0.1 to 0.3M  | \$0.2 to 0.5M  | Low    | Commissioning split (Nomad vs owner) written into the test plan                                                                                                                                                                                                          |
| B1 | Station 496: relay program out; RTU, RTAC and transfer trip only (Engineering, 5 Oct 2026)                     | F         | \$6.1 to 6.6M  | \$6.1 to 6.6M  | Medium | Protection-basis memo signed by System Protection (existing SEL-551 and REF610 relays plus an RTAC transfer trip meet the interconnection requirement); RTU, RTAC, Axion and fiber scope priced (no line in Rev 2); asset-condition relays, if any, to the asset program |
| B2 | Two feeders instead of four: two skids, two metering sets, two duct-bank runs (Engineering, 5 Oct 2026)        | C/D       | \$1.0 to 1.2M  | \$1.5 to 1.7M  | Medium | Hosting-capacity and protection study at 5 MW per feeder; N-1 position accepted in writing; D-Line 24211 to two tie-ins; four GSUs unchanged                                                                                                                             |
| B3 | Platform and retaining wall sized after geotech, not before                                                    | E         | \$0.3 to 0.8M  | \$0.4 to 1.2M  | Low    | Geotech report; MassDEP SW45 post-closure permit conditions; NFPA 855 spacing                                                                                                                                                                                            |
| B4 | Battery installation labor from a crew build-up instead of 25% of a 2023 total; fire suppression bought twice? | B         | \$0.5 to 0.7M  | \$0.7 to 1.1M  | Medium | Crew build-up: crane set, anchoring, DC and AC terminations per Voyager unit; confirm whether the Voyager enclosure carries its own suppression (\$132K carried)                                                                                                         |
| B5 | Disconnect switches and PT cutouts priced as separate lines although the Siemens skids carry them              | D         | \$0.3M         | \$0.4M         | High   | Strike lines 07.03.02.04, 06.02.09.00 and the PT cutouts once Siemens confirms the skid scope in writing                                                                                                                                                                 |
| B6 | GSU installation hours: 524 man-hours per pad-mount; count to be settled (four carried, twelve mentioned)      | C         | \$0.2 to 0.2M  | \$0.3 to 0.3M  | Medium | Topology and GSU count in writing; crew build-up for a pad-mount set; 2026 transformer quote with the 29-70 week lead time in the schedule                                                                                                                               |
| B7 | Switchyard lighting: 12 HPS luminaires and \$200 a foot of hookup on an unmanned one-acre pad                  | D         | \$0.2M         | \$0.2 to 0.3M  | Medium | Lighting design to the security need (fence-line LED, motion); hookup length and rate from the layout                                                                                                                                                                    |
| B8 | Siting and project management: external siting at \$1.05M and a \$157K siting line inside Station 496          | H/F       | \$0.4 to 0.6M  | \$0.5 to 0.8M  | Low    | Task budget from outside counsel for Article 80 small-project review and the ZBA permit; confirm what the Station 496 siting application is for                                                                                                                          |
| C1 | Section 48E investment tax credit on the storage property                                                      | rev. req. | \$8.4 to 12.4M | \$8.4 to 12.4M | Medium | Eligible-basis opinion; FEOC cost-ratio test (65% non-PFE for a 2028 construction start); normalization election; DPU treatment                                                                                                                                          |
| C2 | Energy-community adder if the closed landfill qualifies as a brownfield                                        | rev. req. | \$2.8 to 4.1M  | \$2.8 to 4.1M  | Low    | Phase II ESA or a prior brownfield-program assessment (the Phase I safe harbor stops at 5 MW)                                                                                                                                                                            |
| C3 | AFUDC: pay the battery at delivery, not a year ahead; protect the siting float                                 | M         | \$0.3 to 0.7M  | \$0.3 to 0.7M  | Medium | Milestone payment schedule in the Nomad SOV aligned to 2028 delivery; float target on the ZBA chain                                                                                                                                                                      |

\*Loaded at the marginal loader of 1.44 (contingency, basis-driven indirects and AFUDC scale with directs; the risk register and booked actuals do not), not the 1.685 average used on slides 5 and 25. Tier C items reduce the revenue requirement, not the \$57.2M. Tiers A and B are gross of the Haugland reconciliation (red team KS1, +\$4M to +\$8M direct), which must close first.

EstimatingCoE | Eversource Energy | Stewards of Affordability

# What the levers are worth, loaded, and how much of it survives the red team's reconciliation



| Tier                 | Loaded range      | Against             |
|----------------------|-------------------|---------------------|
| A. Estimate hygiene  | \$5.0 to 7.1M     | the \$57.2M         |
| B. Value engineering | \$10.1 to 12.3M   | the \$57.2M         |
| A + B                | \$15.1 to 19.5M   | the \$57.2M         |
| Haugland gap (KS1)   | +\$4 to 8M direct | the \$57.2M         |
| C. ITC, adder, AFUDC | \$11.6 to 17.2M   | revenue requirement |

- **Net of the red team:** Tiers A and B (\$15.1 to 19.5M) and the Haugland gap (+\$4 to 8M) are the same order of magnitude. The honest planning case is a \$57M number that holds, not one that falls.
- **The three to pursue now, in order:** A1 and A2 (a decision and a re-wording, about \$3M), B1 (two phone calls to EMA for relay actuals), and C1 (a memo to Tax). Everything else waits on geotech, the hosting-capacity study or the HEG map.
- **What not to do:** take Tier B into the PAF before the studies close. A VE deduct booked ahead of its study is how estimates go to \$74M.

# The investment tax credit is worth \$8.4 to 12.4M to customers and nobody on the project owns it yet

### \$8.4M to \$12.4M

30% credit, eligible basis

Basis \$28M (battery, install, BOP, loaded) to \$41M (plus station, site, testing).

### +\$2.8M to \$4.1M

Energy-community adder

Only if the closed landfill qualifies as a brownfield. Not assumed.

### 65%

Non-PFE cost ratio required

Storage is exempt from the 2027 wind and solar cliff. For a 2028 construction start, 65% of direct cost must come from outside prohibited foreign entities.

| What must close                                                                                                                                                                                      | Owner                   | Why it matters                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|-----------------------------------|
| Eligible-basis opinion: which of layers A to G are energy storage property; interconnection property (Station 496) is outside the storage definition in 48E                                          | Tax                     | Sets the \$28M to \$41M range     |
| FEOC cost-ratio test (Notice 2026-15): cells are 52% of equipment cost in the IRS table, so Chinese cells alone can fail a 65% threshold                                                             | Tax + Sourcing + Nomad  | A failed test is a zero credit    |
| Normalization election for storage (IRA amendment to section 50(d)(2)) and the DPU's treatment of the credit; no Massachusetts precedent yet (Outer Cape and Nantucket pre-date storage eligibility) | Regulatory              | Decides how fast customers see it |
| Prevailing wage and apprenticeship written into the construction contracts                                                                                                                           | Sourcing + Construction | 6% vs 30%                         |
| Brownfield status: a closed municipal landfill is not excluded, but above 5 MW the Phase I safe harbor is unavailable; needs a Phase II ESA or a prior program assessment                            | Environmental + Tax     | The 10-point adder                |

- **Not an estimate change.** The \$57.2M stays; the credit reduces the revenue requirement the customer pays over the asset's life. It belongs in the PAF narrative and the rate case, not in the ESF.
- **Scale:** the low end of the credit is larger than Tiers A and B combined, and larger than the Haugland gap.
- **Risk:** the FEOC rules are the single question that can take it to zero. Ask Nomad for the cell and module origin now, before the final-drawings milestone.
- **Confidence:** medium on the mechanism (law-firm summaries of the July 2025 law and Notice 2026-15; statute not read first-hand), low on the dollars until Tax opines.

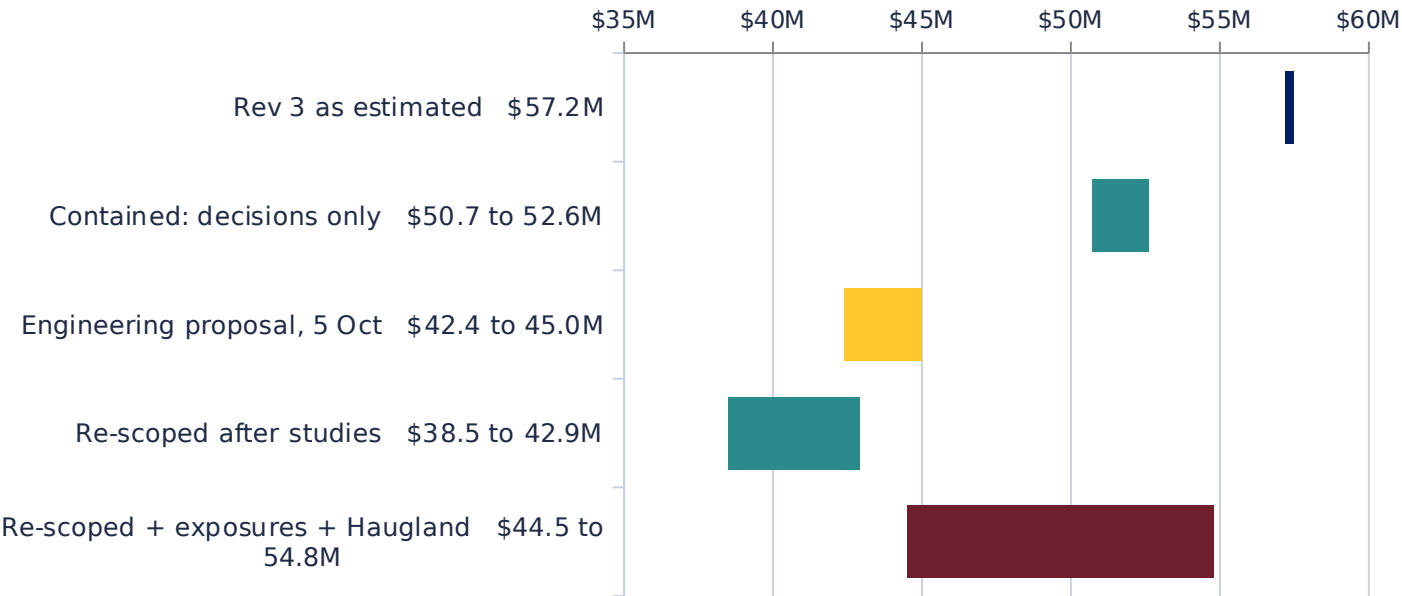
# Scope challenge, layers A to E: a double count on the skids, a GSU count nobody has written down, and civil rates with no basis

| Layer                     | Rev 3 carries                                                                                                                                                                                                                                                                                        | The function requires                                                                                                                                                                                                  | Challenge                                                                                                                                                                                                                                                                                                                                                                                                                             | Direct \$              |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| A. Battery contract       | Nomad fixed price \$12.1M (12 Voyager units, 10-yr warranty \$0.80M, spares, support, shipping, training) plus \$1.13M escalation on the balance                                                                                                                                                     | The battery. Price is fixed until Final Acceptance.                                                                                                                                                                    | Remove the escalation; align Attachment C-2 milestones to the January 2028 delivery; write the tariff and change-in-law clause into the change-order process; treat warranty years 3-10 as a service agreement if Accounting allows                                                                                                                                                                                                   | \$1.1M                 |
| B. Battery installation   | 5,557 mh of install labor (25% of a 2023 site total), 12 pads, fire suppression \$132K, GSU-to-battery conduit and cable                                                                                                                                                                             | Set, anchor and terminate 12 trailer-mounted units; Nomad supervises (two people, one day per container).                                                                                                              | Crew build-up at 200-250 mh per unit (2,400-3,000 mh); confirm whether the Voyager enclosure carries its own suppression before buying \$132K of it twice                                                                                                                                                                                                                                                                             | \$0.5 to 0.7M          |
| C. GSUs and auxiliaries   | 4 x 3 MVA pad-mount GSUs \$1,077K furnish (2023 bidder list + 3%/yr) and \$363K install (2,098 mh = 524 per unit); comms \$355K; AC/DC panels \$137K                                                                                                                                                 | Step 480 V to 13.8 kV for 10 MW. The count follows the topology: one GSU per feeder connection or one per unit.                                                                                                        | Settle the count in writing. The PM's 9 Sep email speaks of twelve GSUs; Rev 3 carries four; Haugland priced the larger count (\$632K deduct item). Install at 524 mh per pad-mount is three to four times a normal set; re-quote the transformer with the 2026 lead time (29-70 weeks) in the schedule                                                                                                                               | \$0.2 to 0.2M          |
| C. GSUs (exposure)        | Four GSUs                                                                                                                                                                                                                                                                                            | If the topology needs one GSU per Voyager unit, twelve                                                                                                                                                                 | Eight more 1 MVA-class pad-mounts, foundations and cable                                                                                                                                                                                                                                                                                                                                                                              | exposure \$0.6 to 1.0M |
| D. Switching station      | 4 Siemens breaker skids \$1,327K furnish (Siemens +/- 30% budgetary at \$300K each incl. relays, cabinets, DC battery, PTs, disconnects); 8 disconnect switches \$238K (1,040 mh); 4 metering transclosures \$265K; lighting \$313K (1,272 mh); duct bank, 15 kV cable, riser poles, SSVT, grounding | Four feeder connections at 2.5 MW each because a 15 kV-class feeder takes about 9 MVA aggregate (National Grid MA limit; Eversource's not retrieved). Each connection needs a breaker, protection, metering and a tie. | (1) The 8 disconnects and the PT cutouts are on the skids per the project team and are priced again as separate lines: \$259K and 1,040 mh. (2) Lighting: 12 HPS luminaires and \$200 a foot of direct-bury hookup on an unmanned one-acre pad; LED on the fence line at a third of the cost. (3) Hosting-capacity study: if two feeders can take 5 MW each, two skids, two GSUs and two metering sets go away (savings register B2). | \$0.4M                 |
| D. Two feeders (exposure) | Four feeder connections at 2.5 MW each                                                                                                                                                                                                                                                               | If two feeders carry 5 MW each: voltage regulation and protection coordination on those feeders, and one feeder outage strands 5 MW rather than 2.5 MW                                                                 | The hosting-capacity study may require voltage mitigation or regulator settings on the two feeders; System Planning has to accept the N-1 position for the Hyde Park load pocket in writing                                                                                                                                                                                                                                           | exposure \$0.2 to 0.5M |
| E. Site and civil         | Cut and re-place 2,520 CY \$456K (2,520 mh); 14-ft retaining wall 280 lf \$470K; trap rock 4,900 sy \$492K; gravel drive 760 lf \$319K; 36 borings \$361K; mob/demob \$313K                                                                                                                          | A stable, drained, accessible pad on a capped construction-debris landfill under a MassDEP post-closure permit; the wall, the platform depth and the cut volume are design outcomes of the geotech, not inputs.        | Unit rates run two to four times library norms (cut at \$181/CY, surfacing \$100/sy, drive \$420/lf, borings \$10K each) on placeholder quantities (2.5-ft platform). No basis is written for either. The reconciliation with Haugland decides which way this layer moves; do not book a deduct before the geotech.                                                                                                                   | \$0.3 to 0.8M          |
| E. Site (exposures)       | No sound mitigation, no visual screening or faux facade, no police details beyond \$12K of traffic control, no NFPA 855 hazard mitigation analysis line                                                                                                                                              | Constructability review: visual screening 'likely required' and a sound study to come; Boston public-way work; NFPA 855 for a lithium system next to housing                                                           | E-23-373 carried \$467K of sound mitigation; Rev 3 dropped it and the risk register does not carry it. Screening and a facade were deferred to 'vendor design'. Price them or register them.                                                                                                                                                                                                                                          | exposure \$0.6 to 1.2M |

# Scope challenge, layers F to M: the relay program, the hours carried from 2023, the siting budget and the loaders

| Layer                            | Rev 3 carries                                                                                                                                                                                                                                   | The function requires                                                                                                                                                                                                                                                                                                              | Challenge                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Direct \$     |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| F. Station 496 feeder protection | 20 relays: hardware \$91K; install 6,400 mh \$1.23M; test 7,800 mh \$1.81M; E&S basis \$1.46M; siting application \$157K; AFUDC, contingency, risk. No line for the RTU, the RTAC/Axon or the transfer trip that the Rev 2 clarifications name. | A transfer trip from the Station 496 feeder breakers to the BESS over the dedicated dark fiber, SCADA points at 496, and feeder protection that tolerates the inverter-based injection on each feeder. Whether that needs 20 new relays, or the existing SEL-551 and REF610 relays with an RTAC, is the protection-basis question. | Engineering proposal (5 Oct 2026): remove the relay program; install the RTU, RTAC and transfer trip only. Net \$6.1M to \$6.6M after a \$0.30M to \$0.55M direct allowance (\$0.67M to \$1.14M at the 496 loader) for the scope Rev 2 names but does not price. Needs a protection-basis memo signed by System Protection; if the legacy relays are an asset-condition replacement, the cost transfers to that program and is not a saving to customers. | \$6.1 to 6.6M |
| G. Testing and commissioning     | Field test 3,200 mh, LCE 1,600 mh, SCADA 1,650 mh, telecom 120 mh: \$1.26M, hours carried from E-23-373                                                                                                                                         | Owner witness and acceptance testing; Nomad commissions its units (contract: \$210K support, \$246K vendor labor)                                                                                                                                                                                                                  | Write the split into the test plan; the owner's hours were set in 2023 before the vendor's scope existed                                                                                                                                                                                                                                                                                                                                                  | \$0.1 to 0.3M |
| H. Owner soft costs              | Siting external \$1,052K plus internal and actuals; PM (contractor and Eversource, three lines plus actuals) \$1.30M; CM \$356K; engineering \$501K contractor plus actuals; outreach, environmental, legal                                     | Article 80 Small Project Review and a ZBA conditional use permit; a PM and a CM for the construction window; 30%-to-IFC engineering                                                                                                                                                                                                | Siting at \$1,052K external is two to three times a typical Article 80 small-project budget; ask counsel for the task budget. PM hours are duration-driven: every month of siting slip adds about \$0.4M across PM, AFUDC and indirects, so the float target on the ZBA chain is the lever, not the rate.                                                                                                                                                 | \$0.2 to 0.4M |
| J. Risk register                 | \$5.3M Sta 360 + \$0.77M Sta 496: material \$1.0M, labor \$0.8M, design changes \$0.8M, UG obstructions \$0.45M, environmental, P&C and station engineering \$0.3M each, soil, weather, outreach, siting, BAR deductibles                       | Risks that the directs and contingency do not already fund                                                                                                                                                                                                                                                                         | Labor-cost and material-cost lines restate escalation (savings register A2); design-change lines overlap the 17.9% contingency; sound mitigation and screening are the risks that are missing                                                                                                                                                                                                                                                             | \$1.3 to 1.8M |
| L. Indirects                     | \$8.06M: E&S station basis \$4.79M (16% of directs, applied to the vendor contract), general services, MDEC, vehicles, non-productive, AS&E, payroll                                                                                            | Template basis rates                                                                                                                                                                                                                                                                                                               | Whether the E&S basis should load a vendor-furnished, vendor-commissioned contract is a policy question (savings register A4, up to \$2.1M); it is the single largest line nobody on the project can change                                                                                                                                                                                                                                               | \$1.0 to 2.1M |
| M. AFUDC                         | \$5.0M over a seven-year development window                                                                                                                                                                                                     | Capitalized carrying cost on CWIP                                                                                                                                                                                                                                                                                                  | Pay the battery at delivery, not a year ahead (C3); protect the siting float; every month costs about \$0.4M all-in                                                                                                                                                                                                                                                                                                                                       | \$0.3 to 0.7M |

# Three numbers: \$57.2M estimated, \$50.7 to 52.6M contained, \$38.5 to 42.9M re-scoped; honest range \$44.5 to 54.8M



- **Read it as a range of outcomes, not a menu.** The contained number needs only decisions and is worth taking to the November PAF. The re-scoped number is a ceiling on reductions; nobody should present it until the geotech, the hosting-capacity study and the Haugland map have closed.
- **What this says about the last-BESS question.** Even fully re-scoped, the station, the platform and the siting case stay in the number: they are the price of this site, not of batteries. A 10 MW battery inside an existing station fence on a standard design would carry neither. If the company builds another, that is the specification.
- **The ITC sits outside all four bars.** \$8.4 to 12.4M of revenue requirement, larger than any case's spread, and the FEOC answer from Nomad decides it.

| Component                              | Loaded \$M        | Needs                                          |
|----------------------------------------|-------------------|------------------------------------------------|
| Reductions in the register             | \$12.9 to 17.1M   | decisions (A, J, L, M) and studies (B to H)    |
| Two-feeder topology (B2), counted once | \$1.5 to 1.7M     | hosting-capacity study                         |
| Exposures not carried                  | \$2.0 to 3.9M     | sound, screening, details, NFPA 855, GSU count |
| Haugland reconciliation                | +\$4 to 8M direct | signed scope map                               |

# Containment plan: five gates, a station-direct design-to-cost target of \$1,250 per kWh, and a monthly number someone owns

| When                                                               | What                                                                                                                                                                                                                                                                                          | Who                                                | What it decides                                                                                                                               |
|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Now to Nov 2026 PAF                                                | Decide A1 (battery escalation), re-word A2, get the Accounting ruling on the E&S basis, strike the disconnect and PT lines that the skids already carry, price or register sound mitigation and screening, open the 48E memo with Tax                                                         | Cost Estimating, PM, Accounting, Tax               | Estimate hygiene: about \$3M to \$5M loaded; the exposures added honestly                                                                     |
| Gate 1, design basis (Q1 2027)                                     | Geotech report, NFPA 855 hazard mitigation analysis, hosting-capacity and protection study for two feeders at 5 MW each, the Station 496 protection-basis memo (RTU, RTAC and transfer trip without the relay program), GSU count and topology in writing, Haugland-to-Rev 3 scope map signed | Substation Eng., Civil, System Planning, HEG       | Decides the two-feeder case (\$1.5M to \$1.7M) and the Station 496 case (\$6.1M to \$6.6M), the civil layer (either way) and the GSU exposure |
| Gate 2, design-to-cost target (before IFC, mid-2027)               | Set a station-direct target of \$1,250 per kWh (\$25M direct, the reference-class 75th percentile) and a loaded target of \$50M; monthly trend review against it; change control board with the CoE; no scope added without a funded source                                                   | PM (accountable), Cost Estimating (trend), sponsor | Turns the estimate into a managed number instead of a forecast                                                                                |
| Gate 3, commercial strategy (before construction pricing, Q3 2027) | Competitive lump-sum pricing for station, civil and install (Haugland plus two), owner-furnished GSUs and skids on alliance pricing, Nomad milestones aligned to delivery, change-order cap, outage windows at 496 priced explicitly                                                          | Sourcing, PM, Construction                         | Converts unit-library construction into a bid; the red team's +25% construction stress test becomes a contract term                           |
| Standing                                                           | Monthly cost-containment report to the sponsor: station-direct \$/kWh against target, forecast at completion, open scope questions, exposures register; the 48E position and the FEOC cell-origin answer tracked as milestones                                                                | PM, CoE                                            | What leadership sees every month until in-service                                                                                             |

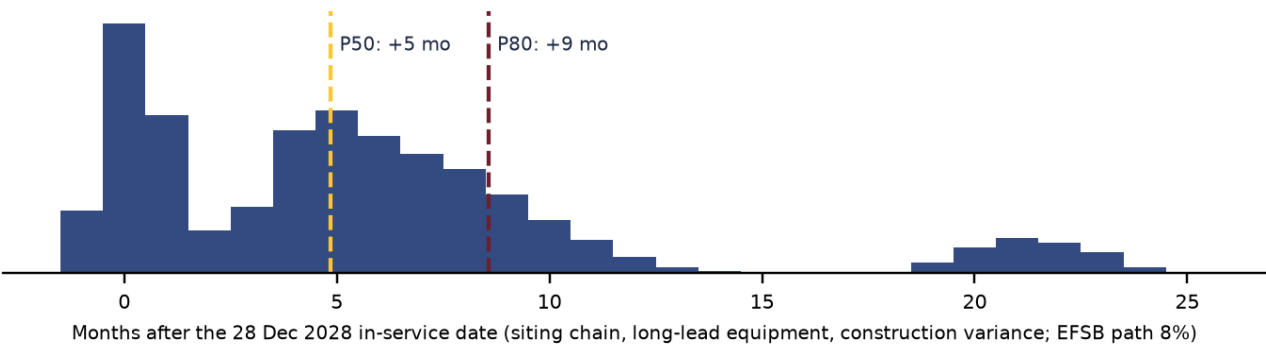
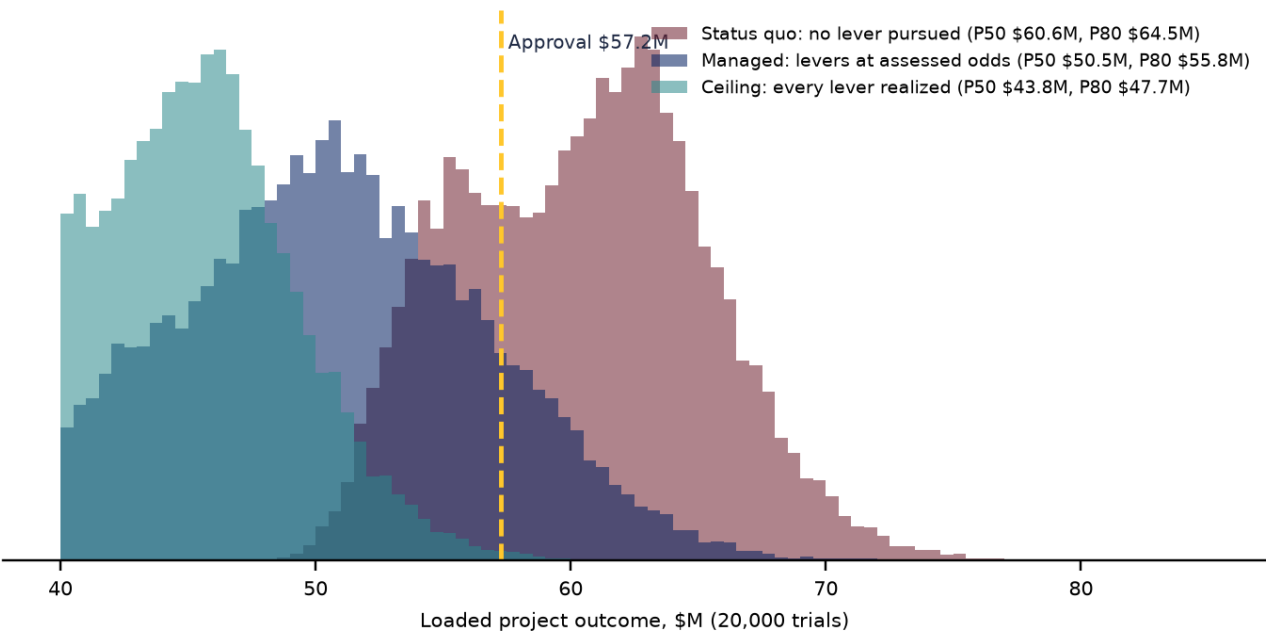
■ **The target is the discipline.** \$25M station-direct is the reference-class 75th percentile for a project this size; the estimate is at \$29.7M. The gap is the scope challenge above, and the PM owns it from the November PAF onward, with the CoE reporting the trend monthly.

# Where to be aggressive: two things drive 54% of the variance, and neither is a line item

| # | Be aggressive on                                                                      | Why, from the simulation                                                                                                                                                                                                                                             | Owner                               | By       |
|---|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|----------|
| 1 | Close the Haugland reconciliation before the PAF                                      | Threat T1: 58% of the variance in the outcome. A signed scope map turns a +\$3M to +\$12M loaded exposure into a known number; until then every reduction below is gross of it.                                                                                      | PM + Cost Estimating + HEG          | Nov 2026 |
| 2 | Buy siting float                                                                      | Schedule: 14% of the variance; P50 +5 months, P80 +9 months, each month about \$0.37M. File Article 80 early, set the ZBA hearing within 90 days, pre-brief the abutters on screening and sound.                                                                     | Siting + PM                         | Q1 2027  |
| 3 | Take the decision-only levers now                                                     | A1 escalation (\$1.3M expected), A2 risk wording (\$1.1M), B5 disconnect double count (\$0.3M), C3 AFUDC timing (\$0.3M): about \$3M expected, no study needed.                                                                                                      | Cost Estimating + Sourcing          | Nov 2026 |
| 4 | Close the Station 496 protection basis: RTU, RTAC and transfer trip, no relay program | Engineering's 5 Oct proposal is the largest single lever in the register (\$6.1M to \$6.6M net) and carries about a quarter of the simulated variance on its own; it needs a signed protection-basis memo and a priced RTU/RTAC scope, neither of which exists today | P&C Engineering + System Protection | Dec 2026 |
| 5 | Settle the topology and the GSU count in one study                                    | B2 (\$0.6M expected) and T6 (\$0.4M expected threat) are the same question asked twice. Hosting capacity for two 5 MW feeders decides both.                                                                                                                          | System Planning + Substation Eng.   | Gate 1   |
| 6 | Hold the civil layer open until geotech                                               | Accuracy on E is 6% of variance and asymmetric (0.6x to 2.6x). B3 is worth \$0.3M expected; the downside is three times that.                                                                                                                                        | Civil + Capital Projects Eng.       | Gate 1   |
| 7 | Get the overhead-basis ruling                                                         | A4 (\$0.5M expected at 30% odds, \$2.1M if granted) is the single largest line nobody on the project can change.                                                                                                                                                     | Accounting                          | Nov 2026 |
| 8 | Price what is missing before someone else does                                        | T2 to T5 (sound, screening, police details, NFPA 855) are \$1.1M expected together; carrying them is cheaper than defending their absence.                                                                                                                           | Siting + Civil                      | Nov 2026 |

Expected values are probability-weighted means from 20,000 trials (mc\_scope\_schedule.py); 'variance' is the Spearman contribution to the spread of the managed outcome. Reductions in the register total \$15.1 to 19.5M loaded if every one lands; the simulation expects about \$10.8M of it at the assessed odds.

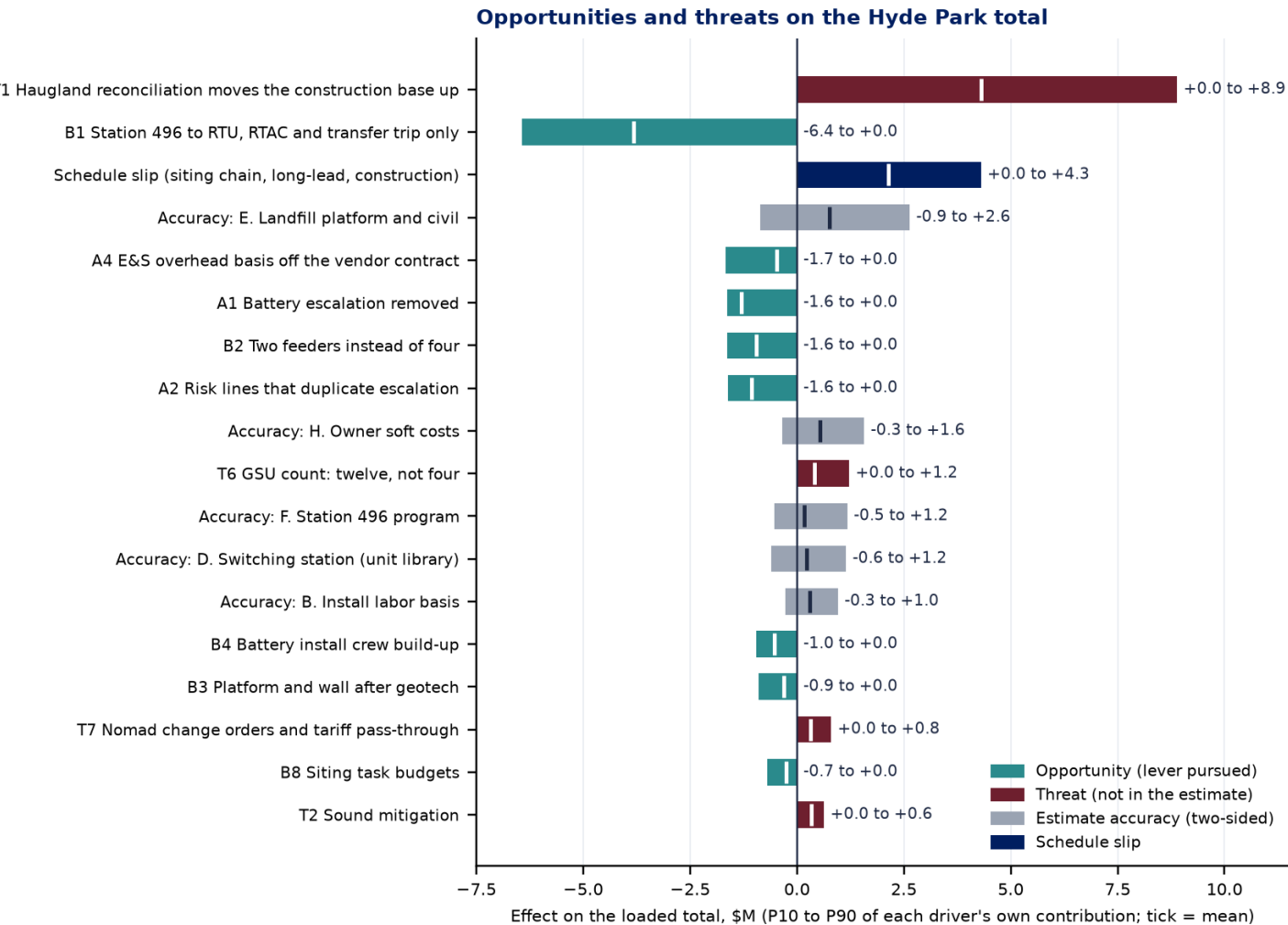
# Monte Carlo on cost, scope and schedule: \$61M if nothing is pursued, \$50M if the levers are worked, \$44M if every one lands



| Posture    | P50     | P80     | P(over \$57.2M) |
|------------|---------|---------|-----------------|
| Status quo | \$60.6M | \$64.5M | 70%             |
| Managed    | \$50.5M | \$55.8M | 14%             |
| Ceiling    | \$43.8M | \$47.7M | 0%              |
| Schedule   | +5 mo   | +9 mo   | 39% over 6 mo   |

- **What the three postures mean.** Status quo: the \$57.2M holds in 30% of trials. Managed: 86%, with a 47% chance of landing under \$50M. Ceiling: every lever lands; it is the floor of what is physically possible, not a plan.
- **The reserve.** The \$7.6M of risk register and contingency is what these outcomes draw on. It covers the managed case at about P75 and the status quo at about P25.
- **Schedule.** Eight in ten trials slip; the median slip is 5 months and costs about \$1.8M; an EFSB path (8%) adds 18 to 24 months.

# Tornado: the Haugland gap and the siting schedule are the risks; the levers are many, small and mostly ours to take



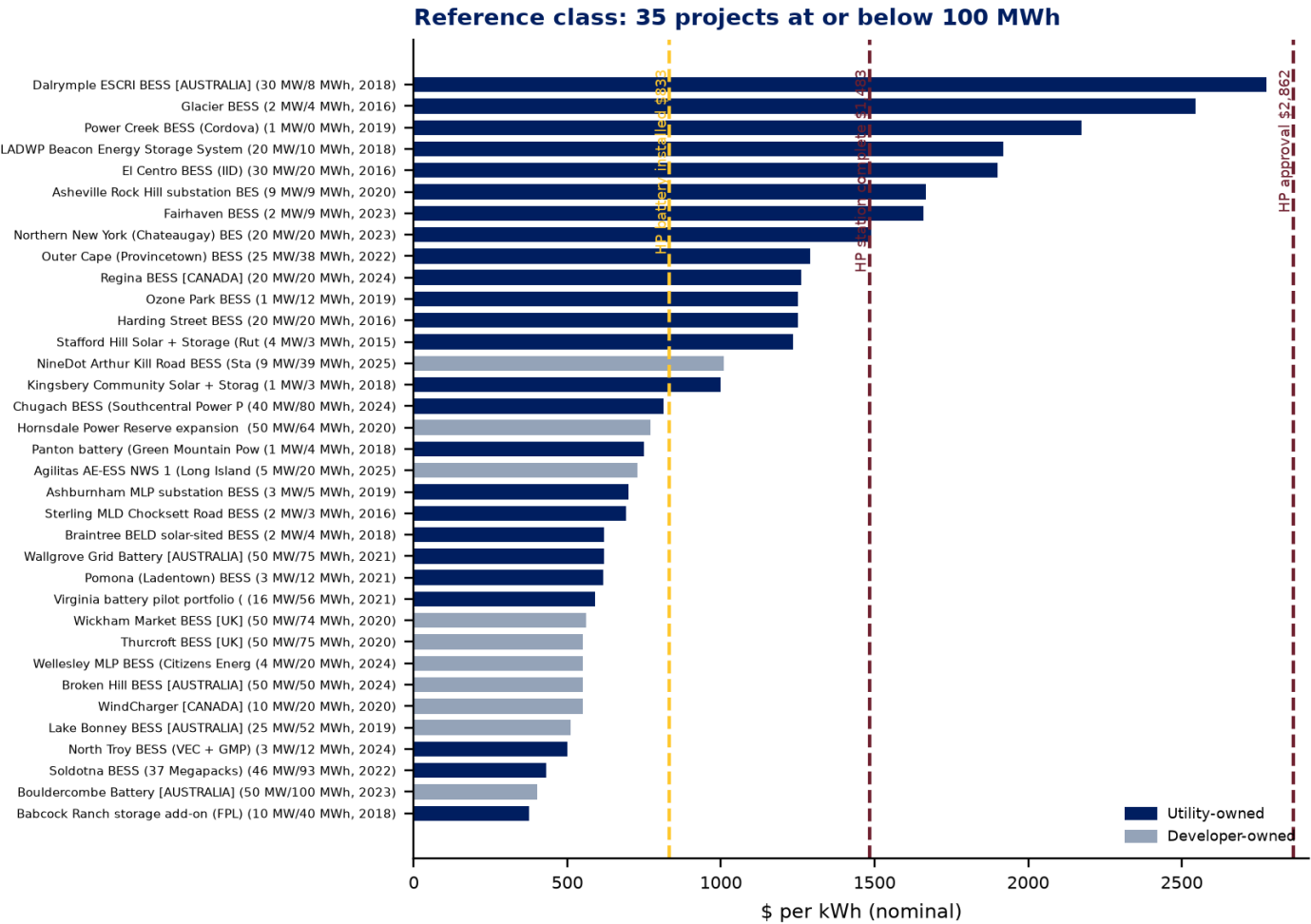
- **Read the right side first.** One threat, the Haugland reconciliation, reaches \$9M at P90 and carries 43% of the variance. Schedule slip reaches \$4.3M. The civil layer and Station 496 are the two accuracy ranges with real upside risk.
- **The left side is a portfolio.** No single lever exceeds \$2M, but together they are worth about \$10.8M at the odds assessed and \$16.8M if every one lands. The four that need only a decision (A1, A2, B5, C3) are the surest.
- **What this says about being aggressive.** Aggression on the station lines buys \$3M to \$6M; aggression on the reconciliation and the siting calendar protects \$8M to \$13M. Do both, in that order.
- **Not on the chart:** the Section 48E credit (\$8.4 to 12.4M), which is larger than everything on the left combined and sits in the revenue requirement.

# What must close before this number goes to EPAC as full funding

| Item                                                                                                                                                                                                                                                   | Type                     | Dollars at stake                                              | Owner                               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|---------------------------------------------------------------|-------------------------------------|
| Reconcile Haugland Class 3 base (\$19.2M, its scope) to Rev 3 layers B+D+E (\$8.3M direct). Red team expects it to move the base up, not down.                                                                                                         | Scope boundary           | +\$4M to +\$8M direct if half the gap is 21334 scope          | Cost Estimating + PM + HEG          |
| Siting chain has no float: ZBA determination Feb 2028, construction start Feb 2028, ISD Dec 2028; EFSB fallback 18-24 mo                                                                                                                               | Schedule                 | ~\$5M per 12 months (AFUDC, escalation, PM, indirects)        | Siting + PM                         |
| Present full funding at its real maturity: 30% engineering, no bid, placeholder platform = Class 4                                                                                                                                                     | Classification           | Range is -25/+50% (\$43M-\$86M) until closed                  | Cost Estimating                     |
| Geotechnical investigation and NFPA 855 hazard mitigation analysis before civil freeze                                                                                                                                                                 | Study                    | \$1.0M of deducts vs \$0.6M-\$3.5M exposure                   | Civil + Capital Projects Eng.       |
| Confirm four written design-basis items (retaining wall, basin, duct bank, GSU count)                                                                                                                                                                  | Design confirmation      | \$1.47M HEG deduct ceiling (~\$0.8M net of GSUs already at 4) | Substation Engineering              |
| Escalation on the fixed-price battery balance; check Exhibit C for tariff / change-in-law pass-through                                                                                                                                                 | Decision                 | \$1.13M                                                       | PM / Cost Estimating / Sourcing     |
| Battery installation labor: replace 25%-of-16,801 mh with a crew build-up                                                                                                                                                                              | Estimate                 | +/- \$0.5M (Rev 2 used 50%)                                   | Cost Estimating + Nomad + HEG       |
| Station 496: Engineering proposes RTU, RTAC and transfer trip only, no relay program (5 Oct); needs a protection-basis memo and a priced RTU/RTAC scope (not in Rev 2); if the legacy relays are asset-condition, the cost transfers rather than saves | Scope / protection basis | \$6.1M to \$6.6M net of a \$0.7M to \$1.1M RTU/RTAC allowance | P&C Engineering + System Protection |
| Spot generation if the BESS misses Dec 2028 (SSF: 2-8 MW) was removed from scope; who carries it?                                                                                                                                                      | Scope / ownership        | Not in project                                                | System Planning                     |
| Re-base D-Line 24211 (2023 conceptual \$4.8M; HEG take-off larger)                                                                                                                                                                                     | Estimate                 | Unknown                                                       | Distribution Engineering            |

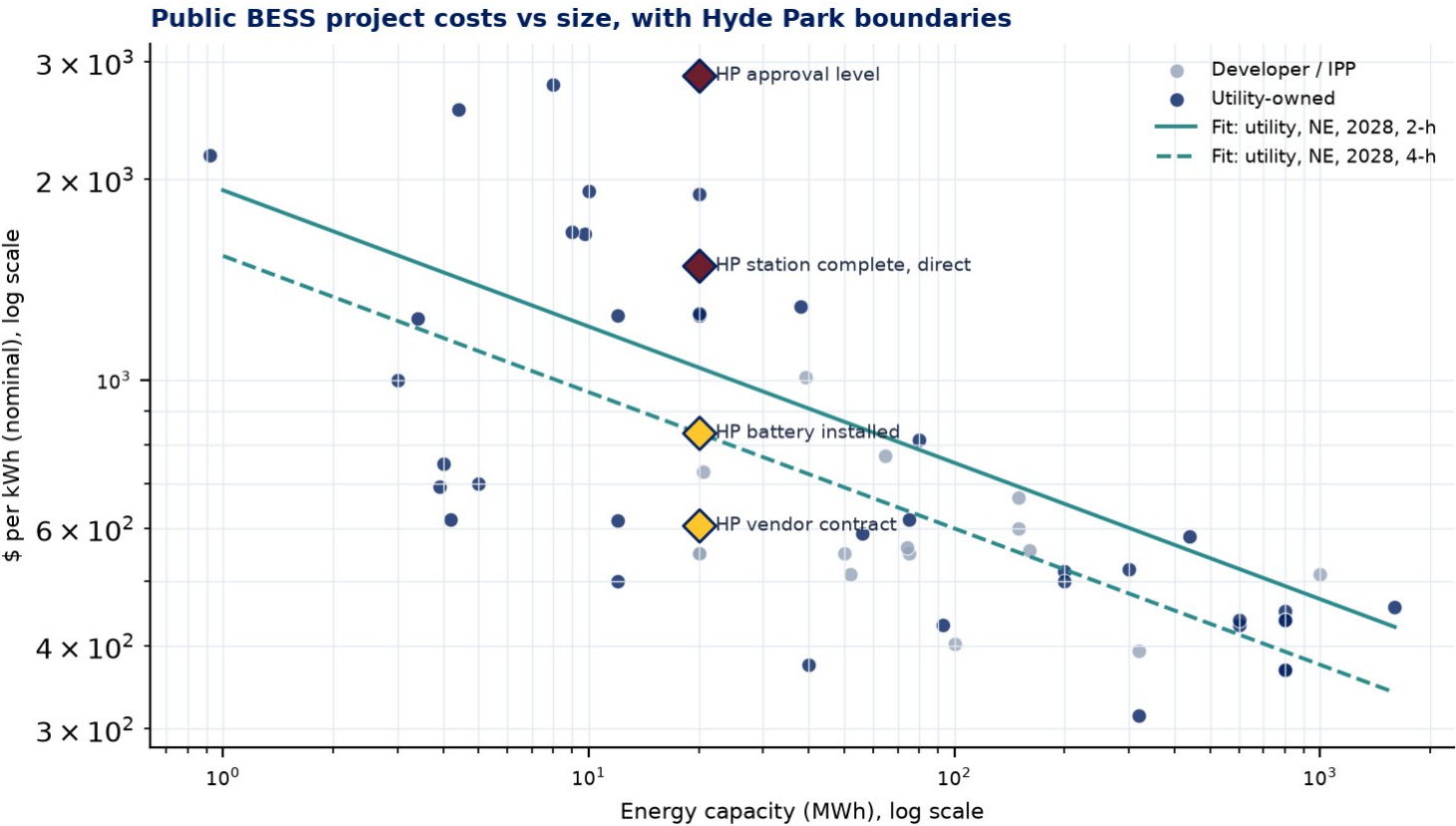
Defensible reductions (~\$3M: battery escalation, design-basis confirmations, material risk sized on non-battery material) are conditional on the Haugland reconciliation closing first.

# 35 built projects at or below 100 MWh, utility- and developer-owned: where Hyde Park sits



- **Median** \$750/kWh; interquartile \$556 to \$1,275. Utility-owned subclass (25): median \$1,235.
- **Hyde Park battery installed, \$833**: 57% percentile.
- **Station complete, direct, \$1,483**: 77% percentile (68% in the utility subclass), beside Fairhaven (\$1,659 actual, 2023), Chateaugay (\$1,490, 2023) and Provincetown (\$1,289 actual, 2022).
- **Approval level, \$2,862**: above every member. The loaders (\$23.7M) and the two interconnection scopes are what the comparables do not carry.
- **Re-based to 2026 dollars (slides 12 and 14 to 18)**: median \$742 at the installed index, \$816 to \$737 across 30 to 50% anchor battery share (2024 basis, back-cast to each project's year).
- **Built projects only (5 Oct 2026)**. Status of every dataset row verified; 27 rows that were planned, under development, under construction or cancelled are excluded from the class and the regression. 14 members carry a final or actual cost (median \$756); the others are contract, budget or announcement figures for projects that were built.
- **Caveat**: figures from regulator, IDA, EIA and utility documents, verified by status audit; developer-owned members carry developer overhead and profit but no utility indirects or AFUDC; incentives are not netted.

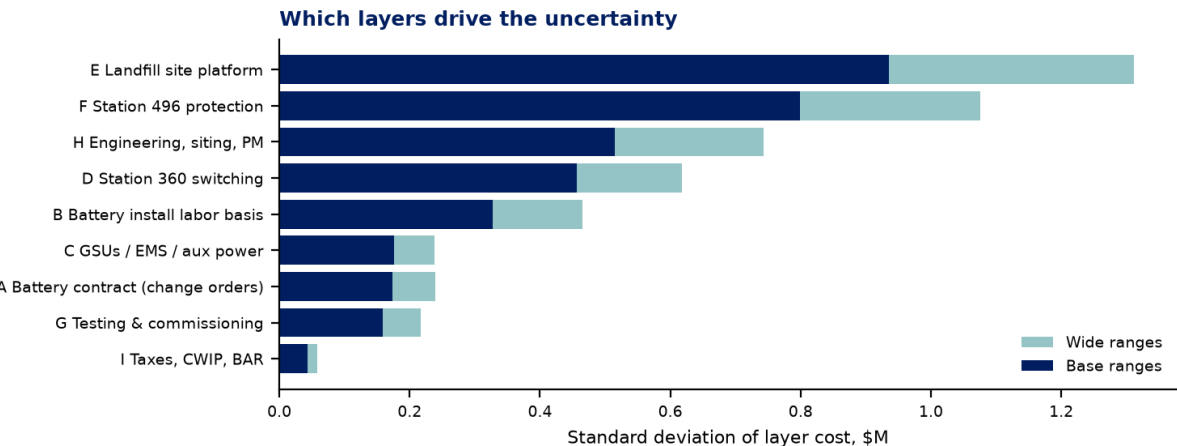
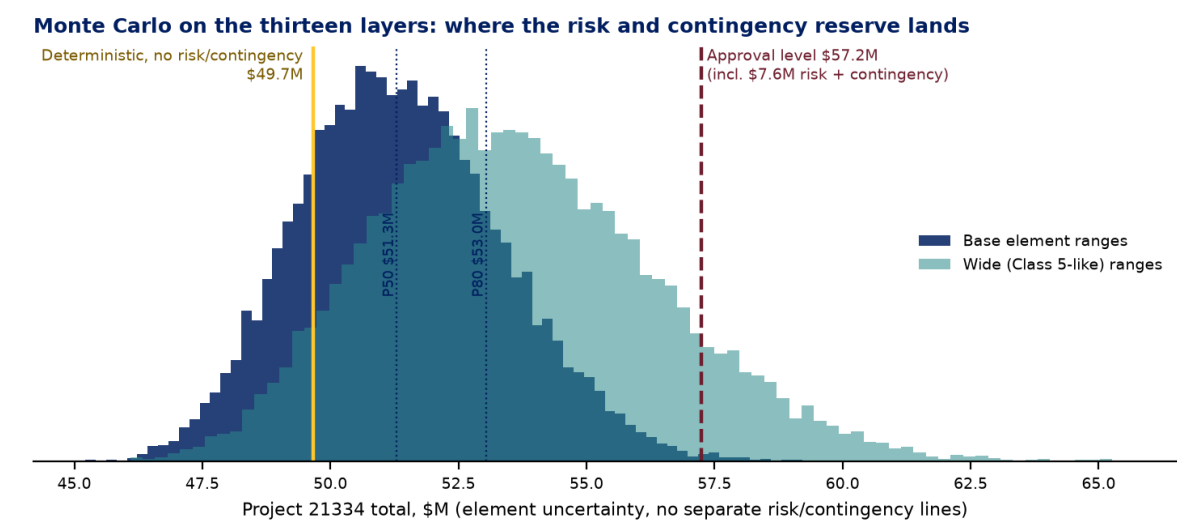
# Size and duration explain half the spread in public costs; Hyde Park's direct cost is high-normal



| Term            | Coefficient | p      |
|-----------------|-------------|--------|
| Size (ln MWh)   | -0.18       | 0.000  |
| Duration (ln h) | -0.34       | 0.003  |
| Year            | +0.017      | 0.55   |
| R-squared       | 0.62        | n = 53 |

- **20 MWh, 2-h, 2028 prediction:** \$968/kWh, 80% interval \$571 to \$1,640.
- **Hyde Park percentiles:** contract 13%, installed 36%, station direct 85%, approval 100%.
- **Use as a bracket, not a price.** Residual spread is a factor of 1.4. In the fuller model utility ownership adds 26% at the same size, duration and year (p 0.01); region is not significant.

# Monte Carlo on the layers: reserve covers the modeled ranges, thin against history



|                            | Base ranges | Wide ranges |
|----------------------------|-------------|-------------|
| P50 total                  | \$51.3M     | \$53.4M     |
| P80 total                  | \$53.0M     | \$55.9M     |
| Reserve to reach P80       | \$3.4M      | \$6.2M      |
| Carried risk + contingency | \$7.6M      | \$7.6M      |
| P(total > \$57.2M)         | 0%          | 10%         |

- Distributions** are PERT ranges on each of the nine Station 360 layers plus Station 496, indirects and AFUDC; risk register and contingency are not stacked on top (they fund the same outcomes).
- Estimate-to-actual uplift on comparables:** Fairhaven +64%, Provincetown +9 to +40%, Waena +37%, Regina +31%, Hyde Park Sep cycle +21%. Median about +31%; Rev 3 carries 17.9%.
- Next step:** replace these ranges with the project team's in a two-hour range-estimating session (AACE 41R-08) before full funding.

## CONTRIBUTIONS

# A \$12 million battery is a \$57 million project, and now every dollar between them has a name

- **Where the money sits.** 292 lines in thirteen layers, reconciled to the penny: battery contract 21%, station and interconnection 32% loaded, owner scope 19%, loaders 41%.
- **Unit cost at every boundary.** \$605/kWh vendor, \$833 installed, \$1,483 station direct, \$2,862 project. Each benchmark read against the row that shares its definition.
- **How it changed.** The battery got cheaper; the project got real. \$43.9M to \$57.2M, five in-service dates, documented from the ESF revisions and the email record.
- **Can it be defended.** Against 35 public projects at or below 100 MWh the station-direct cost is at the 77% percentile and the approval level above every member; a regression on 53 projects brackets a 2028 build at \$571 to \$1,640/kWh; six price curves re-base the class within a tenth of each other. Three kill shots, seven stress tests, a pre-mortem, ten reconciliations.

## Decisions requested

- Adopt the layered view (battery, station, interconnection, owner scope, loaders) as the standard presentation of this project, internally and to regulators.
- Make the Haugland-to-Rev 3 reconciliation, the geotech and the hazard mitigation analysis preconditions of the November PAF; present at Class 4 until they close.
- Assign owners to the siting-chain float and the five early-warning indicators.
- Open the Section 48E credit question with Tax now: eligible basis, the FEOC test on the Voyager cells, and the normalization election.
- Adopt the containment plan: a \$1,250 per kWh station-direct target at IFC, five gates with owners, competitive lump-sum pricing of the station and civil scope, and a monthly cost-containment report.